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Pore-scale modelling of complex conductivity of saturated granular materials

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Pore-scale modelling of complex conductivity of saturated granular materials

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ABSTRACT

The complex conductivity method has been frequently used in solving hydrogeological, engineering, and environmental problems in practice. Macroscopic geophysical responses are governed by pore-scale rock properties; therefore, a clear petrophysical understanding of how the pore-scale structure controls electric conduction and polarization in geomaterials is necessary to fully interpret laboratory-/field-scale geophysical observations. In this study, we have developed a pore-scale numerical approach to calculate the complex conductivity of water-saturated granular materials. This physics-based model has advantage over phenomenological or empirical models such that the intrinsic relations between pore structure and geoelectrical signals could be revealed. In the modelling, the influence of electrical double layer, which is quantified by complex surface conductance, can be converted to an apparent volumetric complex conductivity of either solid particles or pore fluid. The effective complex conductivity of the sample is determined by directly solving the finite-difference representation of the Laplace equation in the domain of a representative elementary volume. The numerical approach is tested on one synthetic sample and one glass beads sample. For the synthetic sample, the consistency between the numerical and the analytical solution confirms the accuracy of the approach. Comparison with experimental measurements of the glass beads sample reported previously indicates that the developed numerical approach can well reproduce the key characteristics of the complex conductivity of water-saturated granular materials in the frequency range between 10^{-1} and 10^{4} Hz. Furthermore, the numerical examples also show that the proposed approach can capture the salinity-dependent electric conduction and polarization in saturated granular materials.

INTRODUCTION

Induced polarization (IP) is a useful geophysical technique for numerous environmental and hydrogeological applications (Kemna *et al.* 2012; Revil *et al.* 2012). In practice, IP measurement can be conducted in either the time domain or the frequency domain. When in the frequency domain, it is usually termed as spectral induced polarization (SIP), which measures the complex conductivity of a material. The real part of complex conductivity quantifies the ease of an electric current passing through the material, whereas the imaginary part of complex conductivity is a measure of the electric energy stored from the external electric field. The complex conductivity of a saturated geological material is controlled by electrical properties of individual phases, i.e., pore fluid and solid particles. The geometric characteristics of individual phases such as porosity, surface area, and particle/pore size also affect the effective complex conductivity of geomaterials (Börner and Schön 1991; Scott and Barker 2003; Revil, Florsch and Camerlynck 2014a).

In addition to solid particles and pore fluid, the solid–liquid interface could also influence the complex conductivity of geological materials at low-to-intermediate frequencies (from mHz to MHz). When a two-phase geomaterial is subjected to an external electric field, charges may accumulate at the solid–fluid interface because of electrical heterogeneity, giving rise to an overall separation between the negative and positive charges. Macroscopically, the material exhibits an electric polarization, termed as interfacial polarization (Hanai 1960) or Maxwell–Wagner polarization. This polarization, which occurs at intermediate frequencies (from kHz to MHz), is usually modelled with the differential effective medium (DEM) theory (e.g., Chen and Or 2006).

At low frequencies (<kHz), the electrical double layer (EDL) forming at the solid–liquid interface could significantly affect both the electric conduction and polarization. In the EDL, the ion density is usually larger than that in the bulk solution (Glover *et al.* 1994), and therefore, the electrical conduction along the liquid–solid interface could be enhanced (Revil and Glover 1998). In addition to electric conduction, the EDL also plays an impor-

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tant role in electric polarization (Revil 2012; Bücker and Hördt 2013). Two prevailing theories were developed to explain the role of EDL in controlling the IP response of geomaterials at low frequencies: EDL polarization (Schwarz 1962) and membrane polarization (Marshall and Madden 1959). Recent developments of these theories have incorporated electrochemical properties of the EDL. To a certain extent, both theories could explain the effect of fluid salinity, pH, temperature, and/or surface charge density on the imaginary conductivity of geological materials (Revil 2012, 2013; Bairlein *et al.* 2016; Hördt *et al.* 2016; Leroy *et al.* 2017). However, it is still unclear how these two models can be reconciled. Therefore, further studies in modelling IP responses of geomaterials are strongly needed.

Recent developments and advances in pore-scale imaging techniques such as X-ray tomography, scanning electron microscopy (SEM), and focused ion beam SEM have enabled the morphological visualisation of the microstructure of geomaterials (Fredrich 1999; Wildenschild and Sheppard 2013). Based on the digital representation of the microstructure, it is possible to compute a geomaterial's physical and transport properties from first principles. Related examples include but not limited to simulation of elastic moduli, electrical resistivity, dielectric constant, water retention curve, and permeability of various geological materials (Saenger *et al.* 2011; Madonna, Almqvist and Saenger 2012; Andrä *et al.* 2013a, b; Blunt *et al.* 2013). Our understanding of the physical and transport properties of porous media has been greatly advanced in recent years with the aid of pore-scale numerical modelling techniques.

However, the pore-scale modelling based on a digital representation of actual microstructures has not yet been applied to study the IP response of geological materials at low-to-intermediate frequencies. It is anticipated that the IP related pore-scale modelling can be used, for instance, to study the controversial controlling pore geometry parameters for the imaginary conductivity (Revil 2014; Weller, Slater and Nordsiek 2014; Niu *et al.* 2016a). It can also be used to test the existing complex conductivity models of geomaterials, such as Archie's model and its modified forms (Glover, Hole and Pous 2000), DEM theory-based models (Bussian 1983), and volume average-based models (Pride 1994; Revil 2013). Another potential application of pore-scale IP modelling is the joint calculation of IP and permeability. The simulated results, along with the digital representation of the material's microstructure, can be used to evaluate some IP-based permeability prediction models (Slater *et al.* 2014; Revil *et al.* 2015).

The objective of this study is to develop a pore-scale modelling technique to simulate the IP response (in terms of complex conductivity) of saturated geomaterials. For simplicity, only granular materials with spherical particles are considered here. We first briefly review the theories describing the influence of EDL on the electric conduction and polarization of geomaterials. The contribution of EDL is quantified by a surface property (complex surface conductance), which is then converted to an apparent volumetric property for solid particle or pore fluid. Then, the effective complex conductivity of the granular material is numerically determined by solving the Laplace equation. Next, the numerical approach is tested on a synthetic sample with a simple cubic structure and on a glass beads sample reported in the work of Leroy *et al.* (2008). In the last section of this paper, we use this approach to analyse the salinity effect on the complex conductivity of water–saturated granular materials.

INFLUENCE OF THE ELECTRICAL DOUBLE LAYER

In soils and rocks, the mineral surface is usually negatively charged because of isomorphous substitutions (McBride 1989) and/or the presence of hydroxyl group (Zhuravlev 2000). When saturated with an electrolyte, ions will accumulate at the solid–liquid interface to balance the negatively charged mineral surface, forming the so-called EDL. In the inner layer of the EDL (i.e., Stern layer), the cations are adsorbed on the mineral surface. The outer layer (i.e., diffuse layer) extends from the outer plane of the Stern layer into the pore space. The electrical properties of the EDL are distinct from the bulk solution, and the EDL can significantly influence the effective electrical properties of the material.

Influence of the diffuse layer

In the EDL, ion density is larger than that in the bulk solution, and thus, electrical conduction can be enhanced at the solid–liquid interface. As discussed in the work of Revil and Glover (1998), both the diffuse layer and Stern layer could contribute to the surface conduction. In addition, the migration of protons on the mineral surface might also contribute to the surface conduction (O'Konski 1960; Skold, Revil and Vaudelet 2011; Leroy *et al.* 2013). In this study, we adopted the assumption used by Revil (2013). That is, the surface conductance at the low-frequency limit C_S^0 is solely from the diffuse layer expressed as

$$C_S^0 = \beta_{(+)} Q_d \,, \tag{1}$$

where Q_d and $\beta_{(+)}$ are the surface charge density and the ion mobility related to the diffuse layer. Note that equation (1) can be easily replaced by other models [e.g., equation (1) in the work of Revil and Glover (1998)] to consider other possible contributions.

The diffuse layer could also contribute to the electric polarization of a porous medium. Currently, there are two different mechanisms quantifying the role of the diffuse layer in the polarization. The first one addresses the influence of normal flow of ions in the diffuse layer on the Stern layer polarization (e.g., Lyklema, Dukhin and Shilov 1983); the second mechanism is related to the variation in ionic concentration in the diffuse layer induced by an external electric field (e.g., Shilov *et al.* 2001). In this study, only the first mechanism is considered and the details will be discussed in the next subsection along with the Stern layer polarization.

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Influence of the Stern layer

The low-frequency IP response of geomaterials was usually interpreted by the theory of Stern layer polarization (e.g., Revil 2012) or membrane polarization (e.g., Bairlein *et al.* 2016). For granular materials characterised by large grain sizes, Stern layer polarization is most likely the controlling mechanism. In this section, we briefly review the key aspects of the Stern layer polarization model. Consider a spherical particle with radius r immersed in NaCl solution. After applying an electric field, the ions in the Stern layer will redistribute due to two driving forces: (1) electric field and (2) ion concentration gradient. The induced complex conductance C_S^s follows a simple Debye relaxation expressed as (Schwarz 1962)

$$C_S^s = \frac{i\omega\tau}{1+i\omega\tau}\Sigma_S^s, \tag{2}$$

where Σ_S^s is the surface conductance associated with the Stern layer, $\omega = 2\pi f$ is the angular frequency (with f being the frequency), and $i^2 = -1$. Parameter Σ_S^s has the following form:

$$\Sigma_S^s = \beta_{(+)}^s Q_s \ , \tag{3}$$

where Q_s and $\beta_{(+)}^s$ are the surface charge density and the ion mobility related to the Stern layer. The relaxation time is related to the radius of the curvature of the EDL r_c by (Schwarz 1962; Niu and Zhang 2017)

$$\tau = \frac{r_c^2}{2D^+}, \tag{4}$$

where D^+ is the diffusion coefficient of the ions in the Stern layer. For granular materials considered in this study, the EDL is coating on the particle surface, and therefore, r_c is equal to the particle radius r_d.

Equation (4) does not consider the polarization of the diffuse layer. To account for the effect, we adopted the diffuse layer polarization model developed by Lyklema *et al.* (1983), and thus, equation (4) can be adapted as (Niu and Revil 2016)

$$\tau = \frac{r_d^2}{2MD^+}. \tag{5}$$

Coefficient M can be related to the electrochemical properties of the EDL by

$$M = 1 - \frac{2}{\cosh\left(\frac{e\varphi_d^0}{2k_bT}\right)}\left[\frac{Q_0\kappa}{4eC_f} - \sinh\left(\frac{e\varphi_d^0}{2k_bT}\right)\right], \tag{6}$$

where $Q_0 = -Q_s - Q_d$ is the total surface charge density, e denotes the elementary charge, T is the absolute temperature, k_b is Boltzmann's constant, and C_f is the electrolyte concentration. The potential at the boundary between the Stern and diffuse layers φ_d^0 can be related to Q_d by the Gouy–Chapman relationship as follows:

$$Q_d = -\sqrt{8\varepsilon_w\varepsilon_0 k_b T C_f}\sinh\left(\frac{e\varphi_d^0}{2k_bT}\right), \tag{7}$$

where ε_w is the relative permittivity of the electrolyte and ε_0 is the vacuum permittivity. The Debye length κ^{-1} can be determined as

$$\kappa^{-1} = \sqrt{\frac{\varepsilon_w\varepsilon_0 k_b T}{2e^2 C_f}}. \tag{8}$$

Upscaling complex surface conductance

By combining the contributions from the diffuse layer and Stern layer, the EDL-induced complex surface conductance C_S^* can be then written as

$$C_S^* = C_S^0 + C_S^s. \tag{9}$$

Note that C_S^* is a surface property of the EDL with the unit in Siemens. Assume that the particle with the EDL is electrically equivalent to a particle without the EDL but with apparent complex volumetric conductivity σ_S^* (in S/m). The surface conductance can be then upscaled as

$$\sigma_S^* = \frac{2C_S^*}{\Pi}, \tag{10}$$

where Π is a characteristic particle size (Niu *et al.* 2016a). For spherical particles, Π is equal to the particle radius r_d (O'Konski 1960; Schwarz 1962). By using equation (10), the surface conductance (in S) can be then upscaled as volumetric conductivity (in S/m) for the solid phase (grains). The term "grain upscaling" is used in this study to represent the above upscaling process, and the word "grain" addresses the solid phase that gains apparent volumetric conductivity σ_S^*.

The complex surface conductance can also be treated as a perturbation of the fluid conductivity, and the increased complex conductivity $\Delta\sigma_w^*$ is then expressed as (Johnson, Koplik and Schwartz 1986; Niu *et al.* 2016a)

$$\Delta\sigma_w^* = \frac{2C_S^*}{\Lambda}, \tag{11}$$

where Λ is a characteristic pore size (Johnson *et al.* 1986), and it can be related to surface-area-to-pore-volume ratio S_{por} and the formation factor F by

$$\frac{2}{\Lambda} = -\frac{\mathrm{d}\ln F}{\mathrm{d}\ln\phi}S_{por}, \tag{12}$$

where ϕ is the porosity. If using Archie's equation $F = \phi^{-m}$ (Archie 1942), equation (12) can be simplified as

$$\frac{2}{\Lambda} = mS_{por}, \tag{13}$$

where m is the cementation factor. Using equation (11), the EDL-induced complex surface conductance (in S) can be then converted to an apparent volumetric (complex) conductivity (in S m^{-1}) for pore fluid. Therefore, the complex conductivity of the fluid phase after upscaling becomes $\sigma_w + \Delta\sigma_w^*$. Similarly, we use the term "pore upscaling" to represent the above upscaling process, and the

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word "pore" addresses the fluid phase that gains apparent volumetric conductivity $\Delta\sigma_w^*$. Note that, in "pore upscaling", only the fluid phase gains additional conductivity while the conductivity of the solid phase is not affected by the EDL.

EFFECTIVE COMPLEX CONDUCTIVITY CALCULATION

In the last section, the EDL-induced complex surface conductance has been converted to an apparent conductivity of the solid particle or a perturbation in the fluid conductivity. Therefore, the geomaterial with solid, liquid, and EDL can be then simplified as a mixture of only solid particles and fluid. In this section, we discuss how to calculate the effective complex conductivity of this two-phase mixture.

Definition of effective complex conductivity in heterogeneous material

Consider the laboratory measurement of the electrical conductivity of an isotropic material. The applied current I flows through a sample, and the resulting potential difference ΔP across the sample is measured. Considering the sample's geometry, the average electrical field ‹**E**› and current density ‹**J**› can be calculated, and then the effective conductivity σ_{eff}^* of the material is determined from

$$\langle \mathbf{J} \rangle = \sigma_{eff}^* \langle \mathbf{E} \rangle . \tag{14}$$

Note that ‹**J**› and ‹**E**› usually are not in phase because of the storage of electric energy in the material, and therefore, the effective conductivity σ_{eff}^* is a complex number. Its in-phase and quadrature components are σ_{eff}' and σ_{eff}'', respectively. In a heterogeneous material, the electric field **E** and the resulting current density **J** are not uniform, and they are functions of the spatial position $\mathbf{x} = (x, y, z)$ where x, y, and z are coordinates. Their relations to ‹**E**› and ‹**J**› are as follows (Mendelson 1975; Cheng and Torquato 1997):

$$\langle \mathbf{E} \rangle = -\frac{1}{V}\int_V \nabla\Psi(\mathbf{x})\mathrm{d}V \tag{15}$$

$$\langle \mathbf{J} \rangle = \frac{1}{V}\int_V \mathbf{J}(\mathbf{x})\mathrm{d}V, \tag{16}$$

where V is the domain of a representative elementary volume of the sample. The local current density $\mathbf{J}(\mathbf{x})$ and potential $\Psi(\mathbf{x})$ in the heterogeneous material can be determined by solving the Laplace equation as follows:

$$\nabla \mathbf{J}(\mathbf{x}) = -\nabla\left[\sigma^*(\mathbf{x})\nabla\psi(\mathbf{x})\right] = 0, \tag{17}$$

where $\sigma^*(\mathbf{x})$ is the local complex conductivity of the sample. If the location **x** is within the pore space, $\sigma^*(\mathbf{x})$ will have the same conductivity value with the pore fluid; otherwise, it has the electrical property of the solid phase.

Numerical calculation of σ_{eff}^*

To calculate the effective conductivity σ_{eff}^* of a heterogeneous material, the first step is to determine the local complex conductivity of the sample. In "grain upscaling", the fluid conductivity is $\sigma_w + \mathrm{i}\varepsilon_w\varepsilon_0\omega$ where ε_w is the relative permittivity of the fluid (the term $i\varepsilon_w\varepsilon_0\omega$ denotes the contribution of dielectric effect), and the apparent conductivity of the solid phase is the sum of the surface conductivity σ_S^* [equation (10)] and the dielectric effect, i.e., $i\varepsilon_S\varepsilon_0\omega$, where ε_S is the relative permittivity of solid particles. In pore upscaling, the conductivity of the solid phase is only arisen from the dielectric effect with a value of $i\varepsilon_S\varepsilon_0\omega$, whereas the fluid has an apparent conductivity of $\sigma_w + \mathrm{i}\varepsilon_w\varepsilon_0\omega + \Delta\sigma_w^*$, the last term of which is calculated with equation (11).

At a given frequency, once the local complex conductivity is defined, a potential gradient can be applied across the sample. The spatial distribution of electric potential Ψ and current density **J** in the sample can be then determined by solving equation (17) with the applied boundary conditions. Afterwards, the volume-averaging electric potential and current density, i.e., ‹**E**› and ‹**J**›, can be determined according to equations (15) and (16). Finally, the effective conductivity of the sample σ_{eff}^* can be calculated with equation (14). It is noted that, since the local conductivity varies with frequency [e.g., see equations (2) and (10)], the effective conductivity of the sample is also frequency dependent.

The analytical solution to equation (17) only exists for heterogeneous materials with very simple microstructures. More generally, the equation has to be solved numerically to obtain the effective complex conductivity of the sample. In this study, we adopted an industrial standard code (AC3D.F) developed in the National Institute of Standards and Technology (available at *http://ciks.cbt.nist.gov/garbocz/manual/node52.html*). The code has been widely used to determine the effective conductivity/permittivity of heterogeneous materials (e.g., Torrents, Mason and Garboczi 2000; Kidner *et al*. 2002). It solves the finite-difference (FD) representation of the Laplace equation with a conjugate gradient method. Periodic boundary conditions are imposed during the calculation (e.g., see Guibert *et al.* 2016). Both isotropic and anisotropic materials can be handled by the code. For details of the algorithm, one can refer to Garboczi (1998). In the calculation, the sample is discretised into a number of cells, and each cell has a uniform electrical property. We modified the code to allow for dynamic allocation of memory, and thus, the model resolution can be increased with a given sample size. We also changed the format of the input files such that the complex conductivity at each cell can be assigned individually for the calculation.

Validation of the numerical calculation

To validate the numerical calculation, a synthetic sample with the simple cubic structure is considered (Figure 1). In each unit element of the lattice structure, there is a sphere in the centre of the cube. Assume that the background conductivity of the sample (i.e., fluid phase) is σ_1^* and the conductivity of the sphere (i.e.,

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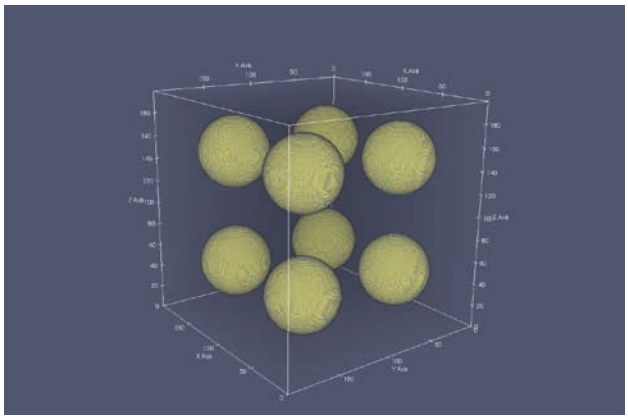


Figure 1 The 3D illustration of the simple cubic structure.

solid phase) is σ_2^* Numerical simulations were performed to calculate the effective conductivity of the sample in Figure 1. A unit element is considered, and the particle radius is 35 μm. In the calculation, we assume that the imaginary conductivity of each phase is only from dielectric polarization. Then, the complex conductivity σ^* of each phase (e.g., solid phase or liquid phase) can be expressed as

$$\sigma^* = \sigma' + i\sigma'' = \sigma' + i\varepsilon^r\varepsilon_0\omega \tag{18}$$

where σ' and σ'' represent the real and imaginary parts of the complex conductivity of the phase, respectively, and ε^r is the related relative permittivity. In the numerical calculation, the fluid conductivity is 1×10^{-4} S m^{-1} and the relative permittivity ε^r of water is 80; the conductivity and relative permittivity of solid phase are 5×10^{-5} S m^{-1} and 4, respectively. Thus, the complex conductivity of fluid phase σ_1^* is equal to 1×10^{-4} S m^{-1} + $i\,80\varepsilon_0(2\pi f)$ S m^{-1}. The complex conductivity of the solid particle σ_2^* is 5×10^{-5} S m^{-1} + $i4\varepsilon_0(2\pi f)$ S m^{-1}.

It was shown in previous research that the cell size (or model resolution) δ might influence the calculation (Zhan *et al.* 2010). Therefore, it is important to choose an appropriate δ value for the calculation. In this study, we determine δ by analysing its effect on the calculated porosity ϕ and surface-area-to-pore-volume ratio S_{por}. For a fixed particle radius of 35 μm, five model resolutions (δ = 10, 5.3, 2.6, 1.3, and 0.7 μm) were used and the calculated ϕ and S_{por} are plotted in Figure 2. It is shown that ϕ is almost constant while δ varies between 10 μm and 0.7 μm. The value of S_{por} continuously increases as δ decreases. However, when δ is smaller than ~2 μm, the change in the calculated S_{por} is insignificant. As a trade-off between accuracy and computation cost, we chose δ = ~1.4 μm for other calculations in this section.

For the effective conductivity of the sample with a simple cubic structure, Torquato (1985) has developed a theoretical model based on a perturbation expansion of the polarization. The model is based on a set of microstructure-dependent parameters, and truncation of the expansion yields an accurate approximation formula for the effective conductivity of the material (e.g., see

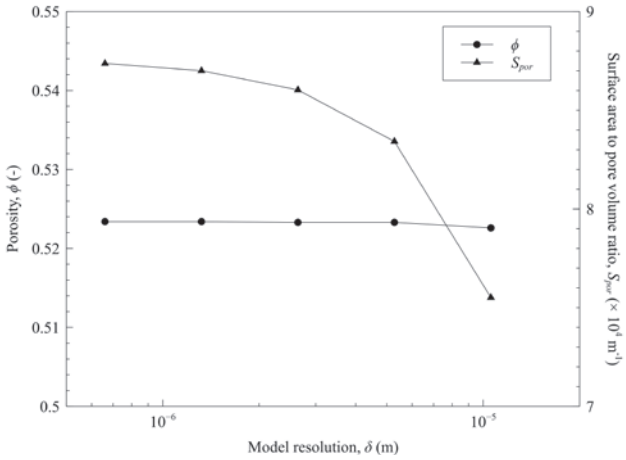


Figure 2 Calculated porosity and surface-area-to-pore-volume ratio of the sample in Figure 1 with different model resolutions.

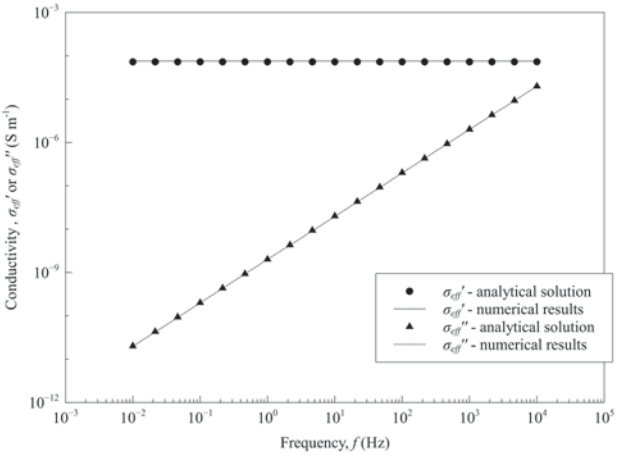


Figure 3 The complex effective conductivity of the sample in Figure 1 at different frequencies.

Cheng and Torquato 1997), which is expressed as (Torquato 1985)

$$\sigma_{eff}^* = \sigma_1^* \frac{1+2f_2\beta_1-2(1-f_2)\zeta_2\beta_1^2}{1-f_2\beta_1-2(1-f_2)\zeta_2\beta_1^2}, \tag{19}$$

where f_2 is the volume fraction of the sphere. The three-point parameter ζ_2 is dependent on the microstructure of the material (e.g., porosity). For the simple cubic lattice structure in Figure 1, the ζ_2 values were reported in Torquato (1985) (see his Table 1). Parameter β_1 is related to the conductivity of the sample, and it has the following form:

$$\beta_1 = \frac{\sigma_2^* - \sigma_1^*}{\sigma_2^* + 2\sigma_1^*}, \tag{20}$$

where σ_1^* and σ_2^* are the complex conductivity values for phases 1 and 2, respectively. Apart from a trivial constant, parameter β_1 is equal to the dipole polarizability of a sphere with conductivity σ_2^* embedded in a matrix of conductivity σ_1^* [e.g., see equation (4) in Sihvola and Kong (1988)].

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The effective complex conductivity of the sample in Figure 1 is calculated both numerically and analytically in the frequency range between 10^{-2} Hz and 10 kHz, and the results are shown in Figure 3. It appears that the complex conductivity calculated from numerical method is in good agreement with the analytical solution in this broad frequency range. To further test the numerical approach, we change the porosity of the sample ϕ by varying the radius but fixing the position of the particle. Same material properties were used, and the numerical results (at 1 Hz) are shown in Figure 4 where the porosity ϕ varies between 0.5 and 0.95. Equation (19) is also used to calculate the complex conductivity of the sample at $\phi = 0.9$, 0.8, 0.7, 0.6, and 0.5. The results are shown in Figure 4 with solid symbols. Again, the numerical solution matches the analytical solution very well, validating the applicability of the numerical approach.

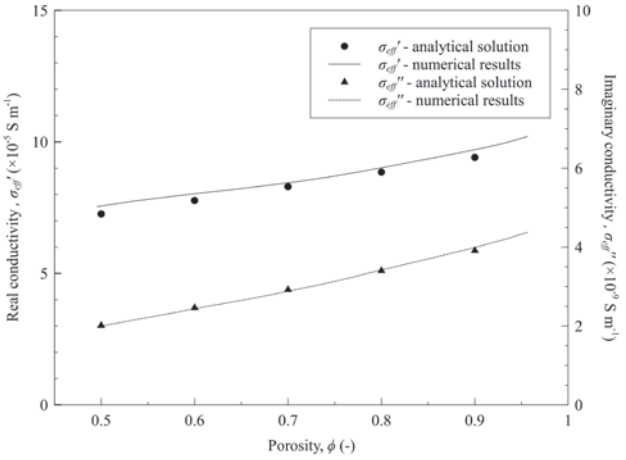


Figure 4 The complex effective conductivity (at 1 Hz) of samples having the simple cubic structure with different porosities.

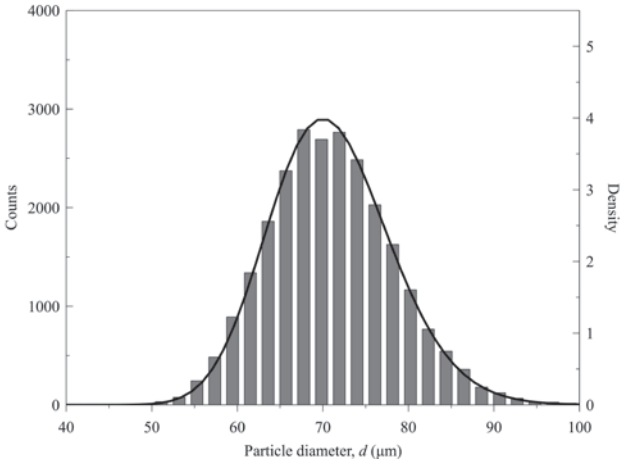


Figure 5 The particle size distribution of the glass beads sample. The solid curve is calculated from lognormal distribution with mean particle diameter d = 70 μm and standard deviation $\overline{\sigma} = 0.1$. The histogram represents the particle size distribution of the numerical sample reconstructed in this study.

COMPARISON WITH EXPERIMENTS

In this section, we test whether the numerical approach can simulate the frequency-dependent IP response of geomaterials. We choose the glass beads sample reported in the work of Leroy *et al.* (2008) to compare with. The sample is numerically reconstructed, and its complex conductivity is then calculated using the aforementioned approach.

Glass beads sample

The sample in Experiment 1 in the work of Leroy *et al.* (2008) is used here. As reported, the particle size of the sample follows a lognormal distribution with mean particle diameter d = 70 μm and the standard deviation $\overline{\sigma} = 0.1$ (Figure 5). The spherical glass beads were packed in a cylinder with 40% porosity and saturated with NaCl solution (σ_w = 0.005 S m^{-1}). In the numerical simulation, we recreate the sample by randomly generating a number of spheres, the radii of which follow the same lognormal distribution (Figure 5). The spheres are packed in a cylindrical container by allowing them to fall in the gravitational field. The total number of the particles in the numerical sample is 16000 and the porosity is 0.39.

Numerical calculation

A cubic sample is taken from the packed cylindrical sample for the calculation of complex conductivity (see a 3D illustration in Figure 6). In the calculation, the fluid conductivity σ_w is assigned as 0.005 S m^{-1}, which is the same as the experimental value. The surface site densities associated with the diffuse and Stern layers are assumed identical, and their values were determined by fitting the calculated curve to experimental data as 10^{16} sites m^{-2} (this corresponds to a surface charge density of 1.6×10^{-3} C m^{-2}). It is noted that the value used here is in the same order with but slightly smaller than those used in the work of Leroy *et al.* (2008), in which 7.5×10^{16} and 6.5×10^{16} sites m^{-2} were used for the diffuse and Stern layers, respectively. The mobility values of Na^+ in the diffuse and Stern layers of the glass beads (silica) are assumed to be the same, i.e., 5.2×10^{-8} m^2 s^{-1} V^{-1} (Revil 2013).

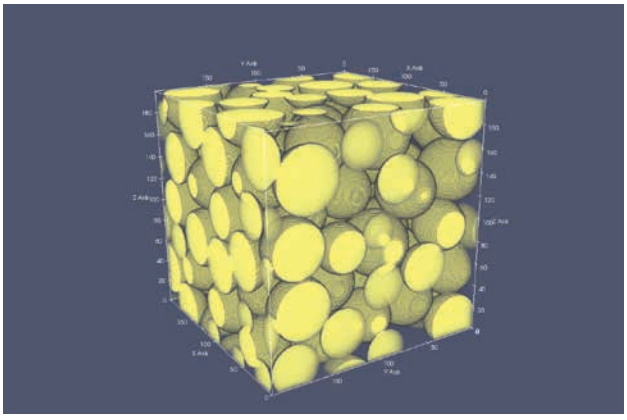


Figure 6 The 3D illustration of the reconstructed sample used in the calculation of the complex conductivity of the glass beads sample. The sample size is 375 μm.

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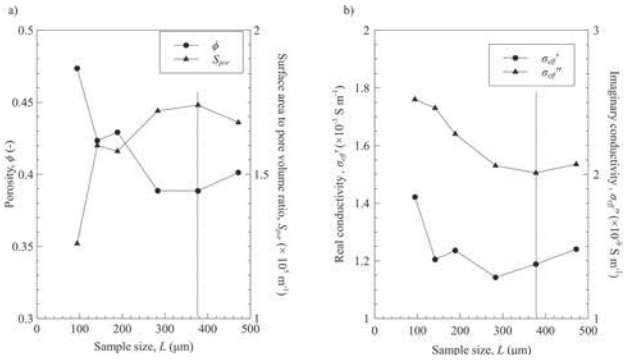


Figure 7 The influence of sample size on the calculated porosity, surface-area-to-pore-volume ratio, and complex conductivity of the reconstructed glass beads sample. The dash line indicates the sample size $L \approx 375$ μm, which was used in the subsequent calculations.

Coefficient M in equation (5) was estimated with equations (6), (7), and (8) by assuming room temperature (20 °C); the estimated $M \approx 2$. The EDL-induced complex surface conduction C_S^* was upscaled using two ways: (1) converting C_S^* to σ_S^* (grain upscaling) and (2) converting C_S^* to $\Delta\sigma_w^*$ (pore upscaling). In either case, the associated phase will have an extra complex conductivity in addition to its intrinsic value as discussed before.

Previous research has shown that the sample size may influence the calculated physical/geometrical properties of geomaterials (Andrä *et al.* 2013b). In general, it is preferred to use a large sample. However, because of the computational limitation, a trade-off has to be made. In this study, we performed numerical calculations with different sample sizes and the trade-off was chosen according to the results (Figure 7). For all the calculations in Figure 7, a fixed model resolution was used, i.e., $\delta = \sim 1.4$ μm. It is obvious that the calculated ϕ, S_{por}, σ_{eff}', and σ_{eff}'' are influenced by the sample size L, as shown in Figure 7. However, when L is larger than 300 μm, it seems that the variations in these calculations are quite small. As a balance between accuracy and computational cost, in the subsequent calculations (see the two subsequent sections), we chose $L = 375$ μm and the cubic sample is discretised into 200^3 cells. This corresponds to a model resolution of ~1.9 μm.

Numerical results and comparison with experiments

The effective complex conductivity of the sample in Figure 6 is calculated in the frequency range between 10^{-1} Hz and 10 kHz, and the calculated phase shift $\theta = \text{atan}(\sigma_{eff}''/\sigma_{eff}')$ is plotted in Figure 8. The measured values of the glass beads sample are also plotted for comparison. In general, the numerical calculations well reproduce the trend of the measured phase spectra at both low (<100 Hz) and intermediate frequencies (>100 Hz). The characteristic frequency (e.g., peak frequency) of numerical curves is slightly smaller than that of the measured spectra. This is probably because the effect of diffuse layer polarization is underestimated in the calculation (i.e., M is underestimated). At frequencies larger than 100 Hz, the calculated phase is slightly

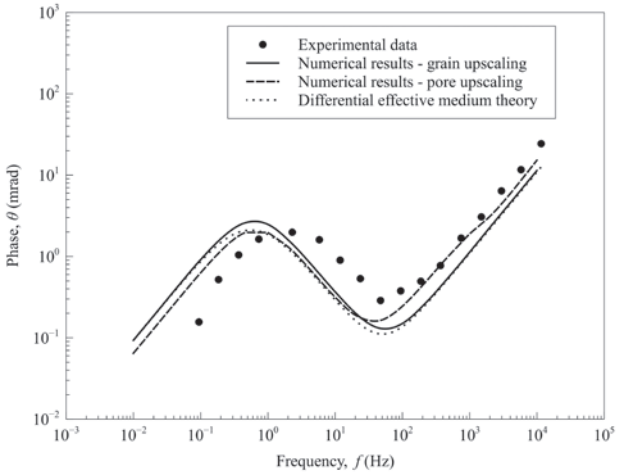


Figure 8 The measured and calculated phase of the glass beads sample. Experimental data are from Leroy *et al.* (2008). In the pore-scale calculation, the sample was numerically reconstructed with spheres with the same size distribution. Theoretical curve from the DEM is also plotted.

smaller than the experimental results. This might be because parasitic capacitance was involved in the experiment (Leroy *et al.* 2008). Grain upscaling and pore upscaling give similar results although noticeable difference can still be found, especially at low frequencies, as shown in Figure 8.

The conductivity model developed by Bussian (1983) is also used here for comparison with the proposed numerical approach. The model is based on the DEM theory, and its complex form is written as

$$\sigma_{eff}^* = \frac{\sigma_1^*}{F}\left(\frac{1-\sigma_2^*/\sigma_1^*}{1-\sigma_2^*/\sigma_{eff}^*}\right)^m, \tag{21}$$

where σ_1^* and σ_2^* are the complex conductivity values of the fluid and grains, respectively. The modelled phase with equation (21) is also plotted in Figure 8 as indicated by the dotted line. In the calculation, parameter m was 1.5 (Leroy *et al.* 2008) and the formation factor F was 3.9 (calculated as $F = \phi^m$). The complex fluid conductivity was identical to the value used in the numerical simulation. The complex conductivity of grains is the sum of σ_S^* [equation (10)] and the dielectric effect $i4\varepsilon_s\varepsilon_0\omega$. In calculating σ_S^*, the electrochemical properties of EDL are same as those used in the above numerical simulations and the characteristic particle size $\prod = 35$ μm. In general, as shown in Figure 8, the phase spectra calculated using the DEM theory are quite similar to the numerical and experimental results, especially for results with grain upscaling.

EFFECT OF FLUID CONDUCTIVITIES

The fluid conductivity can influence the real conductivity of geomaterials. At high salinity, the surface conduction is insignificant; therefore, effective real conductivity σ_{eff}' is linearly proportional to the fluid conductivity (Rhoades, Raats and Prather 1976); the inverse of the slope is known as the intrinsic formation factor F.

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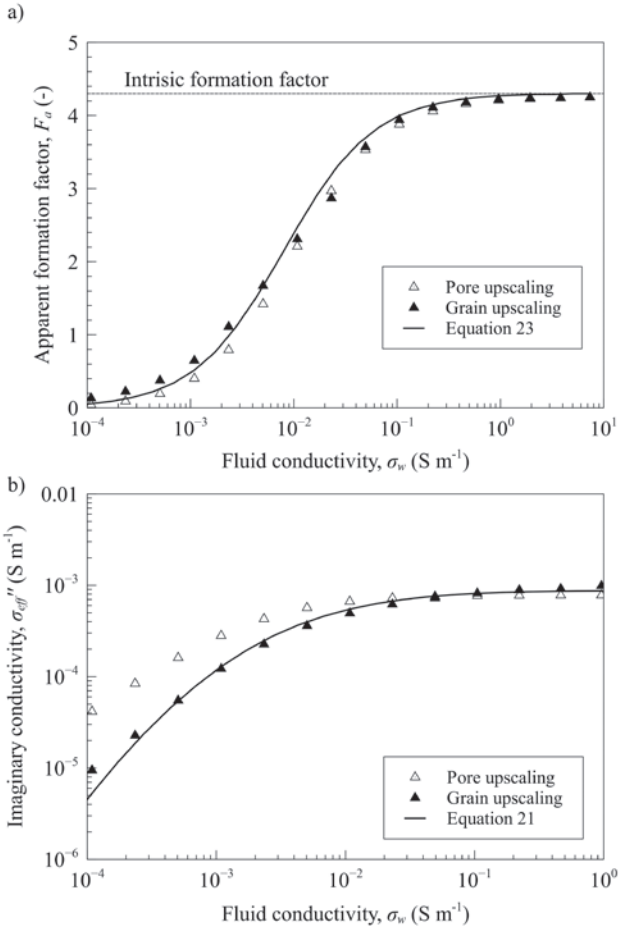


Figure 9 Complex conductivity of the granular sample in Figure 6 at different fluid conductivity values. The data points are from the numerical approach, and solid curves are from the DEM theory.

When the fluid conductivity σ_w decreases, the influence of surface conduction becomes noticeable. The σ'_{eff}–σ_w relation becomes nonlinear, and the ratio σ_w/σ'_{eff} (i.e., apparent formation factor F_a) starts to deviate from the intrinsic formation factor F towards a smaller value. The salinity-dependent F_a has been observed in a range of geomaterials, including clear sandstone (Revil, Kessouri and Torres-Verdín 2014b), clayey sandstones (Niu, Revil and Saidian 2016b), and unconsolidated sediments (Weller, Slater and Nordsiek 2013). The fluid conductivity σ_w can also affect the low-frequency (<100 Hz) imaginary conductivity σ_{eff}'' of a geomaterial (Lesmes and Frye 2001). At low-to-intermediate salinities, σ_{eff}'' increases with σ_w; after reaching a certain salinity, σ_{eff}'' keeps almost constant regardless of the increase in σ_w (Revil and Skold 2011; Weller and Slater 2012).

Numerical calculations

In this section, we use the developed numerical approach to calculate the complex conductivity of the granular material in Figure 6 and test whether it can reproduce the aforementioned trend. In the calculation, the salinity dependence of surface conductance is considered. Instead of solving the equilibrium equations describing the complete chemical reactions on a silica surface in contact with NaCl solution (e.g., Leroy *et al.* 2008), we use a simplified model developed by Revil and Skold (2011) to account for the salinity effect on the surface conductance. The surface conductivity related to the Stern layer is expressed as

$$\Sigma_S^s = e\beta_{(+)}^s \Gamma_S^0 \frac{K_{\mathrm{Na}} C_f}{10^{-\mathrm{pH}} + K_{\mathrm{Na}} C_f}, \tag{22}$$

where Γ_S^0 is the total surface site density and K_{Na} is the equilibrium constant associated with the adsorption of Na^+ at the surface of SiO^- sites on silicate grains. In the calculation, the following parameters were used: $\Gamma_S^0 = 5\times10^{18}$ m^{-2}, $K_{\mathrm{Na}} = 10^{-3.25}$ (Revil and Skold 2011), and pH = 7. Both grain upscaling and pore upscaling were used. The calculated F_a and σ_{eff}'' are shown in Figures 9a and 9b, respectively. In Figure 9a, the intrinsic formation factor F is also plotted, which was numerically calculated by setting the surface conductance as zero.

The DEM-based equation (21) is also used here for comparison, and the results are shown in Figure 9. We defined the ratio of fluid conductivity σ_w and σ_{eff}' as the apparent formation factor F_a. At low frequency, equation (21) can be simplified by ignoring high-order terms expressed as (Niu *et al.* 2016a)

$$F_a = \frac{F\sigma_w}{a + \sigma_w}, \tag{23}$$

where a is equal to $m(1\text{-}1/F)\sigma_S$, with σ_S being the real part of σ_S^*. In the calculation, the following parameters were used: $a = 8\times10^{-3}$ S m^{-1} and $F = 4.3$. Parameter a was determined by fitting equation (23) to the numerical results, as shown in Figure 9a.

Results and discussion

In Figure 9a, the apparent formation factor F_a calculated using the numerical method shows strong salinity dependence. At low salinities, F_a is lower than the intrinsic formation factor F, indicating a significant contribution of the surface conduction. When $\sigma_w > 5\times10^{-1}$ S m^{-1}, the calculated F_a is equal to F, implying the dominance of the bulk conduction. In general, the calculation from the developed numerical approach well reproduces the salinity-dependent real conductivity response of a geomaterial. Moreover, the numerical results fairly match the results from DEM theory, providing additional evidence for the validation of the numerical approach.

Figure 9b presents the imaginary conductivity of the sample. The numerical approach with either grain upscaling or pore upscaling results in salinity-dependent imaginary conductivity, which is in good agreement with trend observed in some published studies (e.g., Revil and Skold 2011; Weller and Slater 2012). The theoretical curve from the DEM theory [equation (21)] is also plotted in the figure, and it agrees well with the numerical results, especially for those calculated with grain upscaling. The consistence between published experimental trends and simulated results again supports the validation of the

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proposed numerical approach. It is also noted that the critical fluid conductivity in Figure 9b (~0.01 S m^{-1}), at which the imaginary conductivity starts to decreases significantly, is roughly one order smaller than that observed in experiments (e.g., ~ 0.1 S m^{-1} in Revil and Skold 2011). The exact reason is unclear, and it might be due to the simplified treatment of the influence of fluid chemistry on surface conductance [i.e., equation (22)].

SUMMARY AND CONCLUSION

In this study, we present a pore-scale numerical approach for the modelling of the complex conductivity σ_{eff}^{*} of water-saturated granular materials. The effect of EDL on σ_{eff}^{*} is considered by assuming that the diffuse layer contributes to the electric conduction whereas the Stern layer is responsible for the electric polarization. The combined effect is quantified by the complex surface conductance, which is then converted to an apparent volumetric complex conductivity for solid particles (i.e., grain upscaling) or pore fluid (pore upscaling). The effective complex conductivity of the sample is determined by directly solving the FD representation of the Laplace equation in a representative elementary volume of the sample. Since the geometrical characteristic and electrical properties of each phase are involved in the simulation, the interfacial polarization can also be accounted for in the numerical modelling.

Numerical simulations are performed on a synthetic sample with a simple cubic structure and a glass beads sample. The calculated complex conductivity of the synthetic sample well matches the analytical solution, showing the accuracy of the developed approach. The pore-scale modelling also reproduces the key characteristics of the IP response of the glass beads sample, including the relaxations at low and high frequencies. These examples provide some evidence for the validation of the numerical simulation.

The developed numerical approach is used to calculate the complex conductivity response of a granular material at different salinities. It is shown that the proposed approach can simulate the salinity-dependent apparent formation factor, which increases with fluid conductivity towards the intrinsic formation factor at high salinities. The imaginary conductivity σ'' calculated from grain upscaling and pore upscaling at different salinities shows the similar trend as observed in published experimental studies. It is also in good agreement with the results from the DEM theory.

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文献 2/4

Joint inversion of NMR and SIP data to estimate pore size distribution of geomaterials

Qifei Niu; Chi Zhang

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Joint inversion of NMR and SIP data to estimate pore size distribution of geomaterials

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Joint inversion of NMR and SIP data to estimate pore size distribution of geomaterials

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SUMMARY

There are growing interests in using geophysical tools to characterize the microstructure of geomaterials because of the non-invasive nature and the applicability in field. In these applications, multiple types of geophysical data sets are usually processed separately, which may be inadequate to constrain the key feature of target variables. Therefore, simultaneous processing of multiple data sets could potentially improve the resolution. In this study, we propose a method to estimate pore size distribution by joint inversion of nuclear magnetic resonance (NMR) T_2 relaxation and spectral induced polarization (SIP) spectra. The petrophysical relation between NMR T_2 relaxation time and SIP relaxation time is incorporated in a nonlinear least squares problem formulation, which is solved using Gauss–Newton method. The joint inversion scheme is applied to a synthetic sample and a Berea sandstone sample. The jointly estimated pore size distributions are very close to the true model and results from other experimental method. Even when the knowledge of the petrophysical models of the sample is incomplete, the joint inversion can still capture the main features of the pore size distribution of the samples, including the general shape and relative peak positions of the distribution curves. It is also found from the numerical example that the surface relaxivity of the sample could be extracted with the joint inversion of NMR and SIP data if the diffusion coefficient of the ions in the electrical double layer is known. Comparing to individual inversions, the joint inversion could improve the resolution of the estimated pore size distribution because of the addition of extra data sets. The proposed approach might constitute a first step towards a comprehensive joint inversion that can extract the full pore geometry information of a geomaterial from NMR and SIP data.

Key words: Electrical properties; Magnetic properties; Microstructure; Hydrogeophysics; Joint inversion.

1 INTRODUCTION

The pore space plays a fundamental role in controlling fluid transport in porous media (Dullien 2012). For geomaterials such as soils and sedimentary rocks, the geometric characteristic of the pore space is usually described by porosity, specific surface area, pore size and its distribution. These parameters have been frequently used in estimating hydraulic properties of geomaterials. For instance, several characteristic pore sizes have been found to be correlated with the saturated permeability of sandstones (Johnson *et al.* 1986; Katz & Thompson 1986); the pore size distribution is an important input for predicting water retention curves (Kosugi 1994; Perrier *et al.* 1996) and unsaturated permeability of soils and rocks (Kosugi 1996; Tuli *et al.* 2001; Vervoort & Cattle 2003). The geometric characteristics of pore space can be directly measured using some imaging methods, for example, X-ray imaging technique (see a review in Wildenschild & Sheppard 2013), scanning electron microscope (e.g. Timur *et al.* 1971) and laser scanning confocal microscopy (e.g. Fredrich 1999). In addition, there are also many indirect methods that can be used to estimate the geometric characteristics of the pore space. One example is the mercury injection capillary porosimetry (MICP), and the measurement is usually interpreted in terms of pore throat size rather than pore size as demonstrated in Dullien (1981); another example is the gas adsorption analysis, which has been frequently used to infer the surface area and pore size of rock samples (e.g. Kuila & Prasad 2013).

The use of geophysical techniques becomes increasingly attractive in estimating the geometric characteristics of the pore space in geomaterials (Tong *et al.* 2006; Mohnke & Yaramanci 2008; Revil *et al.* 2014), mainly due to its non-invasive nature and high spatiotemporal resolution. Another advantage of using geophysical tools is the applicability to field characterizations. The most promising geophysical tools in practice include nuclear magnetic resonance (NMR) and spectral induced polarization (SIP). The traditional NMR method measures the magnetic response of protons under specific sequences of applied magnetic fields (see

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Behroozmand *et al.* 2015 for a review). In an NMR relaxation measurement, the signal magnitude is proportional to the absolute number of protons and thus the volume of hydrogen bearing fluid in the sample. The NMR decay curve can be decomposed to form a distribution of relaxation time (Borgia *et al.* 1998), which is related to surface-to-volume ratio (Brownstein & Tarr 1979), thus the effective pore size (and its distribution) of the material can be estimated. The SIP method measures the electrical response of a material under an external alternating electric field, and the measurements can be expressed in terms of complex conductivity (see Kemna *et al.* 2012 for a review). At low frequencies, the variation in complex conductivity of a geomaterial is mainly controlled by the polarization of the electrical double layer (EDL) forming at the solid–liquid interface (Lesmes & Morgan 2001). The measured complex conductivity spectra can be decomposed to obtain a distribution of relaxation time (Florsch *et al.* 2014). Since the relaxation time is related to the geometry of the EDL (Schwarz 1962), the obtained relaxation time distribution thus can give an estimation for the pore size distribution of the material. For simplicity, in this study we use 'NMR porosimetry' and 'SIP porosimetry' to term the methods of using NMR and SIP data to estimate the pore size distribution of porous media, respectively.

The geometric control of pore space on NMR and SIP responses of geomaterials is quite complicated, and there are still many ongoing studies towards a full understanding of the petrophysical relations (see some recent researches in Keating & Falzone 2013; Müller-Petke *et al.* 2015; Niu *et al.* 2016; Volkmann & Klitzsch 2016). In practice, some simplifications are usually made such that the pore size information of the material can be inferred from the NMR and SIP measurements. For instance, the NMR porosimetry assumes (1) that the magnetic relaxation of protons in porous media is in fast diffusion regime and (2) that the effects of bulk fluid diffusion and pore coupling on NMR relaxation are negligible (see a review in Behroozmand *et al.* 2015). However, it has been found that these assumptions may be violated for materials with large pores and/or large values of surface relaxivity (Grunewald & Knight 2009; Dlubac *et al.* 2014). In SIP porosimetry, it is generally assumed that the complex conductivity of geomaterials at frequencies lower than 100 kHz is solely from the EDL polarization and dielectric polarization (Leroy *et al.* 2008; Revil *et al.* 2014). However, other polarization may also contribute to the measured complex conductivity, such as interfacial polarization (Hanai 1960) and membrane polarization (Bücker & Hördt 2013). These approximate petrophysical relations used in NMR and SIP porosimetries may induce significant errors to the estimated pore size distribution.

Inversion is needed to decompose the NMR relaxation curve and SIP spectra to obtain the relaxation time distribution. As is well known, most geophysical inversions are underdetermined due to the limited number of measurements; moreover, the inversion is also ill-posed because the measurements are usually contaminated by noise (Tarantola & Valette 1982). To mitigate these problems, joint inversion has become a common approach (e.g. Moorkamp *et al.* 2011) because extra information from other data set could reduce the ambiguity or non-uniqueness of the inversion (Haber & Oldenburg 1997; Linde & Doetsch 2016). This has inspired us to consider the joint use of NMR and SIP data in estimating the pore size distribution of geomaterials. One similar attempt in this research direction is the work of Osterman *et al.* (2016) where both NMR and SIP measurements were used to estimate the pore geometric parameters and thus the permeability of sandstones. However, in their study different types of data sets are still processed separately, and a joint inversion of NMR and SIP data is still not realized yet.

In this study, we explore the idea of joint inversion of NMR and SIP data for the estimation of pore size distributions of geomaterials. We first briefly review theoretical backgrounds of NMR and SIP porosimetries with emphasis on commonly applied petrophysical models. We then introduce a joint inversion approach that can simultaneously process NMR and SIP data. In the joint inversion, one aspect of significant importance is the trade-off between different data sets. We propose a two-step method to balance the weight between NMR and SIP measurements. A numerical experiment is designed to demonstrate the applicability of the proposed inversion method. In the last part of the study, we apply this joint inversion to a Berea sandstone sample and discuss the results as well as the limitations of our proposed approach.

2 THEORETICAL BACKGROUND

2.1 NMR relaxation in porous media

In water-saturated geomaterials, NMR relaxation phenomena arises as the result of the interaction of diffusing nuclear spins of protons with their surrounding environment. The NMR relaxation can produce a bulk magnetization, which is governed by the phenomenological Bloch–Torrey equations. Experimentally, the transverse component of the magnetization can be detected using a series of oscillating magnetic field pulses [e.g. the Carr–Purcell–Meiboom–Gill (CPMG) pulse sequence] and the measured signal is in the form of a voltage decay curve, $E_{xy}(t)$. The solution to the diffusion problem can be expressed as the sum of a number of exponential decays (Brownstein & Tarr 1979) and thus the NMR signal can be expressed as

$$E_{xy}\ (t) = E_0 \sum_i f_i e^{-\frac{t}{T_2^i}} \tag{1}$$

where E_{xy} is the transverse component of the NMR signal, E_0 is the initial signal magnitude, f_i is the relative intensities of the magnetic field relaxing with the relaxation time of T_2^i. The initial magnitude E_0 is directly proportional to the number of protons and thus the amount of water in the material and therefore it can be used to estimate the water content of the material. Assuming a uniform magnetic field, the relaxation with time constant ${T_2}^i$ only involves two processes (Brownstein & Tarr 1979) with

$$\frac{1}{T_2^i} = \frac{1}{T_{2B}} + \frac{1}{T_{2S}^i} \tag{2}$$

where relaxation time T_{2B} is related to the bulk fluid diffusion and T_{2S}^i is related to surface relaxation.

The diffusion rate of protons in bulk water in geomaterials is small, $\sim$0.3 s^{-1}; the relaxation can be greatly enhanced at the mineral surface due to the presence of paramagnetic impurities such as Mn^{2+} and Fe^{3+} (Foley *et al.* 1996). Compared with bulk diffusion, the surface relaxation could dominate the relaxation process in a saturated geomaterial such that $\frac{1}{T_{2B}}$ in eq. (2) could be dropped. Assuming no coupling between pores (e.g. see Grunewald & Knight 2009), a relaxation with time constant T_2^i in eq. (1) can be related to a pore by

$$\frac{1}{T_2^i} = \frac{1}{T_{2S}^i} = \frac{1}{\frac{r_i}{\alpha \rho_s} + \frac{r_i^2}{2\alpha D}} \tag{3}$$

where r_i is the effective size of the pore, α describes the shape of the pore ($\alpha = 3$ for spheres), and D is the diffusion coefficient of water. The parameter ρ_s, termed as surface relaxivity, qualifies the efficiency of the surface relaxation. In practice, the term $\frac{r_i^2}{2\alpha D}$ in

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eq. (3) is usually ignored (i.e. assuming in fast diffusion regime), and the relaxation time can be related to the pore size by

$$T_2^i = \frac{r_i}{\alpha \rho_s} . \tag{4}$$

Thus, the relative intensities f_i of the relaxation time T_2^i can be interpreted as pore size distribution. It should be noted that the bulk diffusion term in eq. (2) could be significant if the material is characterized with large pore space and low magnetic susceptibility (e.g. see Grunewald & Knight 2011). In that case, eq. (4) might be insufficient in describing the T_2 relaxation time, and thus the bulk diffusion rate should also be included.

2.2 SIP and EDL polarization in porous media

The SIP measures the complex conductivity response σ^* of a porous medium at frequencies lower than 100 kHz. While the real part of the complex conductivity σ' quantifies the material's ability to conduct an electric current, the imaginary part σ'' reflects the storage capability of electrical energy from the applied electric field. The two parameters are not independent; they are related to each other by the Kramers–Kronig relation (e.g. see Volkmann & Klitzsch 2015). In this study, our discussion will be focused on the imaginary conductivity.

The low-frequency imaginary conductivity of geomaterials is usually interpreted in the framework of EDL polarization. In geomaterials, the mineral surface is usually negatively charged because of isomorphous substitutions (McBride 1989) and/or the presence of hydroxyl group (Zhuravlev 2000). When the material is saturated with an electrolyte (e.g. NaCl solution), some ions (mostly cations) will accumulate at the liquid-solid interface to balance the surface charge, forming the EDL. Consider a spherical pore space saturated with NaCl solution. After applying an electric field, the ions in the EDL will redistribute to gain electric dipole moment. For the spherical EDL, the induced complex conductance follows Debye relaxation (Schwarz 1962),

$$C_i^* = \frac{i\omega\tau}{1 + i\omega\tau} \Sigma_S, \tag{5}$$

where Σ_S is the surface conductance related to Stern layer. The relaxation time constant τ is related to the radius of the curvature of the EDL r_c (see Appendix A for details) by

$$\tau = \frac{{r_c}^2}{2D^+} \tag{6}$$

where D^+ is the diffusion coefficient of the ions in the EDL. Note that eq. (5) does not consider the ion exchange between EDL and bulk solution. For colloids, the EDL is coating the particles, and therefore, r_c is equal the particle radius r_d as shown in Schwarz (1962). For a spherical pore embedded in rock matrix, the radius of the curvature of the EDL r_c is equal to the pore radius r, and therefore the relaxation time τ is

$$\tau = \frac{r^2}{2D^+} . \tag{7}$$

In geomaterials, the shape of the EDL seems to be controlled by the pore space and thus eq. (7) applies. The correlation between SIP relaxation time and pore size have been experimentally observed for a number of geological materials (Binley *et al.* 2005; Kruschwitz *et al.* 2010; Titov *et al.* 2010; Revil *et al.* 2012). For porous media with different pore sizes, the contribution from each pore can be summed up to form the overall response (e.g. see Leroy *et al.* 2008),

$$C_{\mathrm{EDL}}^* = \sum\nolimits_i f_i \frac{i\omega\tau_i}{1 + i\omega\tau_i} \Sigma_S, \tag{8}$$

where C_{EDL}^* is the complex conductance induced by EDL polarization, and f_i is the relative weighting of the pore with size r_i, which corresponds to the relaxation time τ_i.

For geological materials, the (complex) conductance induced by EDL (in S) is usually treated as a perturbation of the pore fluid conductivity (in S m^{-1}) by considering Joule dissipation (Johnson *et al.* 1986; Niu *et al.* 2016). The increased complex conductivity of the pore fluid is sometimes termed as specific surface (complex) conductivity σ_{ss}^*, and it has the following form,

$$\sigma_{ss}^* = \frac{2\left(C_{\mathrm{EDL}}^* + C_S^0\right)}{\Lambda} , \tag{9}$$

where C_S^0 is the DC surface conductance (e.g. see Revil & Glover 1998) and Λ is a characteristic pore size (Johnson *et al.* 1986), which can be related to surface area-to-pore volume S_{por} and formation factor F by

$$\frac{2}{\Lambda} = m S_{\mathrm{por}} \tag{10}$$

where m is the cementation factor. After applying Archie's law, the complex conductivity of the material can be obtained as

$$\sigma^* = \frac{1}{F}\left(\sigma_f + \sigma_{ss}^*\right) \tag{11}$$

where σ_f is the fluid conductivity. Inserting eqs (9) and (10) into eq. (11) yields

$$\sigma^* = \frac{\sigma_f}{F} + \frac{m S_{\mathrm{por}}}{F}\left(\sum\nolimits_i f_i \frac{i\omega\tau_i}{1 + i\omega\tau_i} \Sigma_S + C_S^0\right). \tag{12}$$

In eq. (12), the first term in the right-hand side (a real number) is associated with ionic conduction occurring in the bulk fluid in the pore space. The real part of the second term related to the surface conduction occurring in the EDL (Revil & Glover 1998) and the imaginary part quantifies the EDL polarization. The parameters Σ_S and and C_S^0 can be expressed in terms of fluid chemistry and electrochemical properties of the EDL (e.g. see Revil 2012). In this study, we will not address the physical origin of the petrophysical parameters in eq. (12) such as F, m, Σ_S and C_S^0; instead, we simplify eq. (12) by using several widely used lumped parameters. If the permittivity induced by dielectric polarization $\varepsilon_{\mathrm{inf}}$ is considered, the above equation is reformulated as,

$$\sigma^* = \sigma_\infty - (\sigma_\infty - \sigma_0) \sum\nolimits_i \frac{f_i}{1 + i\omega\tau_i} + i\omega\varepsilon_{\mathrm{inf}}, \tag{13}$$

where σ_∞ and σ_0 are the real conductivities of the sample at low- and high-frequency limits, respectively and their relations to fluid chemistry and EDL properties can be found, for example, in Revil (2012). Since both σ_∞ and σ_0 are optimized during the inversion, to a certain extent, the effects of surface charge density, fluid salinity, and pH are also considered in interpreting the complex conductivity data.

2.3 Petrophysical simplifications in NMR and SIP porosimetries

In this subsection, we summarize the petrophysical assumptions made in NMR and SIP porosimetries. In NMR porosimetry, it is

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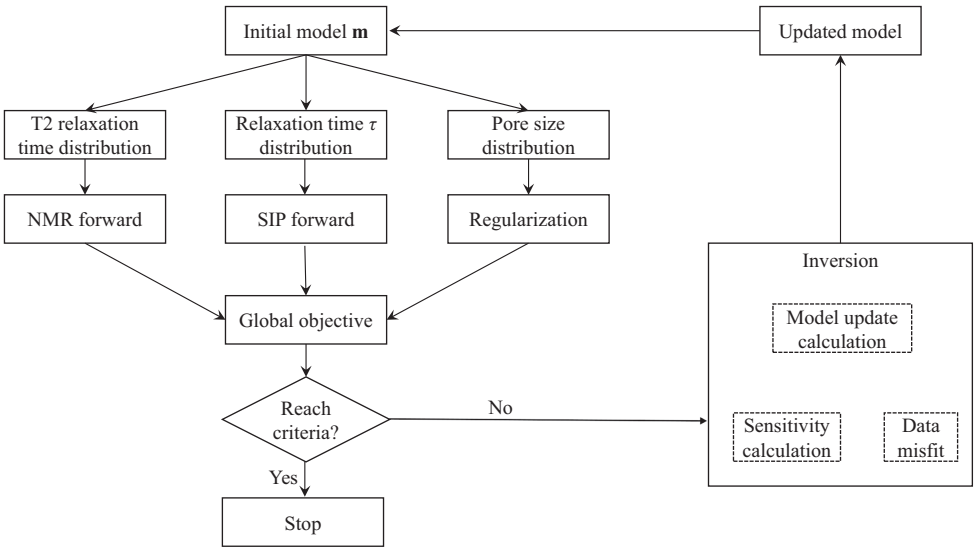


Figure 1. The joint inversion scheme to estimate pore size distribution from NMR and SIP measurements.

assumed that the effect of diffusion on NMR relaxation in a magnetic gradient is negligible. The magnetic gradient arises from the magnetic susceptibility contrast between grain minerals and pore fluid (Cotts *et al.* 1989). This assumption is usually valid if the NMR T_2 relaxation is measured with CPMG pulse sequence with short echo times (Kleinberg & Horsfield 1990). Another assumption in NMR porosimetry is that the pores are isolated and there is no pore coupling effect (D'Orazio *et al.* 1989). For most consolidated sandstones, the narrow pore throat could restrict the molecular diffusion between two interconnected pore spaces during the time of an NMR experiment, and therefore, this assumption is approximately valid (Borgia *et al.* 1996). However, for samples with well-connected heterogeneous pores such as soil aggregates and analogous synthetic silica gels, the assumption might break down (Grunewald & Knight 2009). In addition, the NMR porosimetry assumes that T_2 relaxation is controlled by surface relaxation in the fast-diffusion regime, in which the diffusion-controlled exchange between surface and bulk water is fast enough such that all the water molecules interact with the surface during the NMR measurement (Bryar *et al.* 2000). This fast diffusion regime assumption may not be appropriate for materials with high surface relaxivity and/or large pores (e.g. Keating & Falzone 2013).

In SIP porosimetry, the measurements are usually interpreted with the EDL polarization model. The most important assumption is that the pore size r is equal to the radius of EDL curvature r_c so eq. (7) holds. For isolated pores, this assumption is clearly valid. For geomaterials with interconnected pores, although the correlation between r and τ has been experimentally observed (Revil *et al.* 2012), it remains still unclear whether the relation is quadratic (Scott & Barker 2003; Titov *et al.* 2010). Second, eq. (7) also assumes there is no ion exchange between EDL and bulk fluid (Lyklema *et al.* 1983; Chelidze & Gueguen 1999; Niu & Revil 2016). As a matter of fact, the ion exchange could speed up the ion redistribution in the EDL and thus decrease the relaxation time τ as shown in Lyklema *et al.* (1983). In SIP porosimetry, the interfacial polarization and dielectric effects may also be considered. However, the theory combing EDL polarization, interfacial polarization, and dielectric polarization still significantly underestimates the measured permittivity of sandstones (Lesmes & Morgan 2001; Kremer *et al.* 2016). Lastly, although not stated explicitly, SIP porosimetry applies the isolated pore model, which assumes the ion movement is limited to a single pore or EDL. That is, no ion exchange occurs between two adjacent pores during a SIP measurement.

3 JOINT INVERSION APPROACH

As discussed, NMR and SIP porosimetries involve a number of assumptions about the petrophysical relations of a geomaterial. Since it is difficult to justify these assumptions for a real geomaterial, the pore size information obtained separately from NMR and SIP data may not be accurate. To reduce the uncertainties and thus increase the resolution, in this study we suggest to jointly process the NMR and SIP measurements for the estimation of the pore size distribution of geomaterials. The joint inversion scheme is illustrated in Fig. 1, and each step of the joint inversion is discussed in detail below.

3.1 Petrophysical relations

In practice, there are two types of approaches for joint inversion of two geophysical measurements: the petrophysical approach (e.g. Jardani & Revil 2009) and structural approach (e.g. Gallardo & Meju 2003). The petrophysical approach assumes that different geophysical measurements are functions of the same properties of a material. Thus, multiple geophysical data can be integrated in the inversion. For instance, it assumes both resistivity and seismic velocity of a rock are controlled by its porosity and saturation, and thus resistivity and seismic data can be jointly inverted (Gao *et al.* 2012). Alternatively, the structural approach assumes different geophysical methods yield structurally similar geophysical images of the subsurface. That is, the spatial and directional variations of different geophysical responses are similar. Mathematically, the structural difference between two images can be quantified using the cross-gradient function, and it can be combined with data misfit and smoothness constraints to form the objective function in the joint inversion (Gallardo & Meju 2003; Gallardo & Meju 2004).

In this study, we apply the petrophysical approach in the joint inversion of NMR and SIP measurements. We assume both NMR T_2 relaxation time and SIP relaxation time τ are related to the pore

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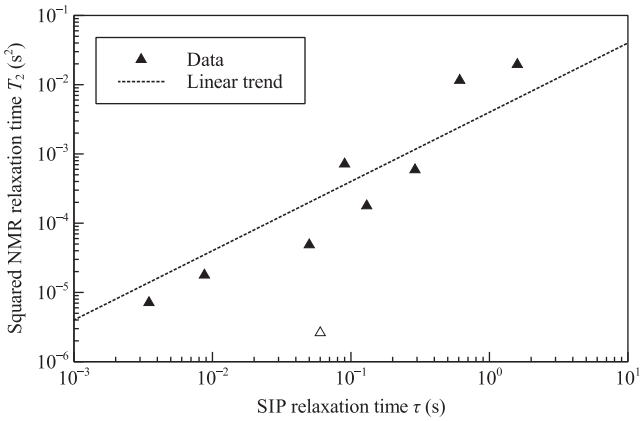


Figure 2. The relation between squared NMR T_2 relaxation time and SIP relaxation time τ of Sherwood sandstone samples. The data are from Osterman *et al.* (2016). The datum with empty triangle is an outlier point.

size r, that is, eqs (4) and (7) are valid. This implies that, for samples with similar surface properties (e.g. D^+ and ρ_s), the relaxation time τ is proportional to squared T_2. We verify the relation between τ and T_2 using Sherwood sandstone samples from Osterman *et al.* (2016). The T_2 squared and τ of these samples are cross-plotted in Fig. 2 where T_2 is the peak NMR relaxation time and τ is determined from the characteristic frequency chosen using the approach in Revil *et al.* (2015). Despite an outliner point, a strong linear correlation is found between ${T_2}^2$ and τ. The experimental evidence lends additional support to the analytical relation between T_2 and τ as presented in the previous section.

3.2 Model parameters

The common relation to pore size r makes it possible to jointly invert the pore size distribution [e.g. the relative intensities/weighting f_i in eqs (1) and (13)] from NMR and SIP measurements. In addition to f_i, other parameters related to the petrophysical models may also be inverted during the inversion, including σ_∞, σ_0, ε_{inf}, and E_0 (e.g. Florsch *et al.* 2014). The surface properties D^+ and ρ_s are also involved in the inversion, but they cannot be recovered in separate inversions because of the non-uniqueness in $\frac{r}{3\rho_s}$ and $\frac{r^2}{2D^+}$, as nicely demonstrated by Mohnke (2014) for ρ_s.

As shown in the previous section and Fig. 2, relaxation time T_2 and τ are related to each other for a given material through the pore size. This implies that the surface properties D^+ and ρ_s are not independent, and therefore it is possible to recover one parameter once the other is known. In fact, previous research has shown, with extra data sets, the joint inversion can properly estimate the material's ρ_s (Mohnke 2014). In this study, we assume D^+ is known as suggested by Revil (2013). The surface relaxivity ρ_s in the inversion can be either assumed as a constant (if we have a prior information on it) or updated iteratively. We define a model vector $\mathbf{m}$, which contains all the unknowns to be recovered from the joint inversion. The model vector $\mathbf{m}$ can be expressed as

$$\mathbf{m} = \begin{pmatrix} \mathbf{x} \\ \ln\sigma_0 \\ \ln\sigma_\infty \\ \ln\varepsilon_{\text{inf}} \\ \ln E_0 \\ \ln\rho_s \end{pmatrix}, \tag{14}$$

where $\mathbf{x}$ is a vector containing the relative weighting f_i for the considered pore size r_i in the form of $x_i = \ln f_i$. The use of log-transformed pore size distribution (relative weighting) and other parameters in eq. (14) is to ensure positivity of the unknowns. Moreover, since the relative weighting f_i of some pores are considerably smaller than the others, the log-transformation of f_i ensures their relative changes are also important during the inversion (Weigand & Kemna 2016).

3.3 Objective function

The objective function Ψ consists of data misfit Ψ_d and a model regularization term Ψ_m (Aster *et al.* 2013),

$$\Psi = \Psi_d + \lambda\Psi_m, \tag{15}$$

where the damping parameter λ controls the relative contribution of the regularization term Ψ_m on the objective function. The regularization is needed to stabilize the inversion process to obtain a unique model (Titterington 1985). In NMR and SIP inversions, Tikhonov regularization is commonly used (Florsch *et al.* 2014; Weigand & Kemna 2016). In this study, we also employ the Tikhonov regularization and the l_2 regularization term Ψ_m can be expressed as

$$\Psi_m = \|\mathbf{W_m}(\mathbf{m} - \mathbf{m}_{\text{ref}})\|_2^2 \tag{16}$$

where $\mathbf{W_m} = \begin{pmatrix} \mathbf{L} & 0 \\ 0 & \mathbf{I} \end{pmatrix}$ is a regularization matrix and $\mathbf{m}$ is a reference model that may contain *a priori* information. While the discrete operator $\mathbf{L}$ in $\mathbf{W_m}$ applies smoothness constraints to the pore size distribution $\mathbf{x}$ in the model vector $\mathbf{m}$, the identify matrix $\mathbf{I}$ forces the petrophysical unknowns (i.e. σ_∞, σ_0, ε_{inf}, E_0 and ρ_s) to be close to reference values defined in $\mathbf{m}_{\text{ref}}$ (a similar treatment can be found in Florsch *et al.* 2014). In this study, the operator $\mathbf{L}$ is chosen as the second-order row difference of an identity matrix (i.e. eq. 19 in Weigand & Kemna 2016).

The minimization of the objective function Ψ can be realized using an iterative Gauss–Newton scheme such that the model update can be determined (e.g. Park & Van 1991) from

$$\left[\mathbf{J}^T\mathbf{W}_d^T\mathbf{W}_d\mathbf{J} + \lambda\mathbf{W}_m^T\mathbf{W}_m\right]\Delta\mathbf{m}_k = \mathbf{J}^T\,\mathbf{W}_d^T\mathbf{W}_d\,(\mathbf{d} - \mathbf{G}(\mathbf{m}_k)) - \lambda\mathbf{W}_m^T\mathbf{W}_m\,(\mathbf{m}_k - \mathbf{m}_{\text{ref}}) \tag{17}$$

where $\mathbf{J}$ is the sensitivity matrix corresponding to the forward modelling function $\mathbf{G}$, which includes both eqs (1) and (13), $\mathbf{W}_d$ is the data weighting matrix, $\Delta\mathbf{m}_k$ is the model update for the model $\mathbf{m}_k$ at the iteration step k, and $\mathbf{d}$ is the data vector. In the joint inversion, the vector $\mathbf{d}$ is expressed as

$$\mathbf{d} = \begin{pmatrix} \mathbf{d}_{\text{SIP}}^{\text{Re}} \\ \mathbf{d}_{\text{SIP}}^{\text{Im}} \\ \mathbf{d}_{\text{NMR}} \end{pmatrix} \tag{18}$$

where vectors $\mathbf{d}_{\text{SIP}}^{\text{Re}}$ and $\mathbf{d}_{\text{SIP}}^{\text{Im}}$ (with the size of $m \times 1$) contain the real part and imaginary part of the frequency dependent complex conductivity, respectively, and the vector $\mathbf{d}_{\text{NMR}}$ contains the measured NMR T_2 relaxation time measurements. Since data points in an NMR experiment is usually at the order of thousands, it will be reduced in the inversion so that the dimension of $\mathbf{d}_{\text{NMR}}$ (with the size of $n \times 1$) is comparable to $\mathbf{d}_{\text{SIP}}^{\text{Re}}$ and $\mathbf{d}_{\text{SIP}}^{\text{Im}}$. The data weighting $\mathbf{W}_d$ matrix contains the errors related to the data vector, and it can be expressed as a diagonal matrix,

$$\mathbf{W}_d = \begin{pmatrix} \beta\boldsymbol{\epsilon}_{\text{SIP}}^{\text{Re}} & 0 & 0 \\ 0 & \beta\boldsymbol{\epsilon}_{\text{SIP}}^{\text{Im}} & 0 \\ 0 & 0 & \boldsymbol{\epsilon}_{\text{NMR}} \end{pmatrix}, \tag{19}$$

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where $\boldsymbol{\epsilon}_{\mathrm{SIP}}^{\mathrm{Re}} = \mathrm{diag}(\frac{1}{\epsilon_j^{Re}})$, $\boldsymbol{\epsilon}_{\mathrm{SIP}}^{\mathrm{Im}} = \mathrm{diag}(\frac{1}{\epsilon_j^{Im}})$, $\boldsymbol{\epsilon}_{\mathrm{NMR}} = \mathrm{diag}(\frac{1}{\epsilon_k})$, and β is a trade-off parameter balancing the weight between SIP and NMR measurements. The parameters ϵ_j^{Re} and ϵ_j^{Im} are the errors related to the real part and imaginary part of the jth complex conductivity measurements, and ϵ_k is the error related to the kth reduced NMR measurement. In practice, these errors can be estimated from the standard deviation of the measurement noise (e.g. Pidlisecky *et al.* 2007).

Starting with a reference model $\mathbf{m}_{\mathrm{ref}}$, the model vector can be updated iteratively until the root-mean-square value of the misfit is smaller than a pre-defined value. At the step k, the model update can be calculated from eq. (17), and thus updated model $\mathbf{m}_{k+1}$ can be determined as

$$\mathbf{m}_{k+1} = \mathbf{m}_k + \chi \Delta \mathbf{m}_k \tag{20}$$

where the step parameter χ belongs to [0, 1]. The restriction of $\chi \leq 1$ prevents the model from overshooting due to non-linearity, and it can be determined using a line search. For details, one can refer to Weigand & Kemna (2016).

3.4 Sensitivity calculation

The sensitivity matrix (also known as Jacobian matrix) $\mathbf{J}$ in eq. (17) contains the partial derivatives of the model responses with respect to the model parameters, expressed as

$$\mathbf{J} = \begin{pmatrix} \frac{\partial \sigma'(\omega_1)}{\partial \ln f_1} & \cdots & \frac{\partial \sigma'(\omega_1)}{\partial \ln f_L} & \frac{\partial \sigma'(\omega_1)}{\partial \ln \sigma_0} & \frac{\partial \sigma'(\omega_1)}{\partial \ln \sigma_\infty} & \frac{\partial \sigma'(\omega_1)}{\partial \ln \varepsilon_{\mathrm{inf}}} & \frac{\partial \sigma'(\omega_1)}{\partial \ln E_0} & \frac{\partial \sigma'(\omega_1)}{\partial \ln \rho_s} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\partial \sigma'(\omega_m)}{\partial \ln f_1} & \cdots & \frac{\partial \sigma'(\omega_m)}{\partial \ln f_L} & \frac{\partial \sigma'(\omega_m)}{\partial \ln \sigma_0} & \frac{\partial \sigma'(\omega_m)}{\partial \ln \sigma_\infty} & \frac{\partial \sigma'(\omega_m)}{\partial \ln \varepsilon_{\mathrm{inf}}} & \frac{\partial \sigma'(\omega_m)}{\partial \ln E_0} & \frac{\partial \sigma'(\omega_m)}{\partial \ln \rho_s} \\ \frac{\partial \sigma''(\omega_1)}{\partial \ln f_1} & \cdots & \frac{\partial \sigma''(\omega_1)}{\partial \ln f_L} & \frac{\partial \sigma''(\omega_1)}{\partial \ln \sigma_0} & \frac{\partial \sigma''(\omega_1)}{\partial \ln \sigma_\infty} & \frac{\partial \sigma''(\omega_1)}{\partial \ln \varepsilon_{\mathrm{inf}}} & \frac{\partial \sigma''(\omega_1)}{\partial \ln E_0} & \frac{\partial \sigma''(\omega_1)}{\partial \ln \rho_s} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\partial \sigma''(\omega_m)}{\partial \ln f_1} & \cdots & \frac{\partial \sigma''(\omega_m)}{\partial \ln f_L} & \frac{\partial \sigma''(\omega_m)}{\partial \ln \sigma_0} & \frac{\partial \sigma''(\omega_m)}{\partial \ln \sigma_\infty} & \frac{\partial \sigma''(\omega_m)}{\partial \ln \varepsilon_{\mathrm{inf}}} & \frac{\partial \sigma''(\omega_m)}{\partial \ln E_0} & \frac{\partial \sigma''(\omega_m)}{\partial \ln \rho_s} \\ \frac{\partial E(t_1)}{\partial \ln f_1} & \cdots & \frac{\partial E(t_1)}{\partial \ln f_L} & \frac{\partial E(t_1)}{\partial \ln \sigma_0} & \frac{\partial E(t_1)}{\partial \ln \sigma_\infty} & \frac{\partial E(t_1)}{\partial \ln \varepsilon_{\mathrm{inf}}} & \frac{\partial E(t_1)}{\partial \ln E_0} & \frac{\partial E(t_1)}{\partial \ln \rho_s} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{\partial E(t_n)}{\partial \ln f_1} & \cdots & \frac{\partial E(t_n)}{\partial \ln f_L} & \frac{\partial E(t_n)}{\partial \ln \sigma_0} & \frac{\partial E(t_n)}{\partial \ln \sigma_\infty} & \frac{\partial E(t_n)}{\partial \ln \varepsilon_{\mathrm{inf}}} & \frac{\partial E(t_n)}{\partial \ln E_0} & \frac{\partial E(t_n)}{\partial \ln \rho_s} \end{pmatrix}, \tag{21}$$

where $\sigma'(\omega_i)$ and $\sigma''(\omega_i)$ are the real and imaginary conductivities of the ith calculated SIP data ($i = 1, 2, \ldots m$) and $E(t_j)$ is the jth calculated NMR data ($j = 1, 2, \ldots n$). The sensitivity can be calculated by several methods (McGillivray & Oldenburg 1990). In this study, we analytically determine each element of the sensitivity matrix $\mathbf{J}$ in eq. (21). For most of the elements in Jacobian matrix $\mathbf{J}$, they are dependent on the model parameters $\mathbf{x}$. To ensure a correct update direction, in the inversion the matrix $\mathbf{J}$ is calculated at each iteration step. If one of the petrophysical model parameters is known (e.g. the surface relaxivity ρ_s is independently measured or estimated), we can set the related sensitivity components (e.g. last column in $\mathbf{J}$ in eq. 21) as zero so that the parameter will not be updated in the inversion.

3.5 Choice of λ and β

The regularization parameter λ is a trade-off between the data misfit and model roughness. For separate inversion, the L-curve method (Hansen 2000) has been frequently used to determine λ. In the joint inversion, the data weighting parameter β balances the relative weighting between SIP and NMR data sets. Similar parameters can also be found in other joint inversions (e.g. Gao *et al.* 2012). When more controlling parameters are involved in the inversion, it is always better to search these two parameters simultaneously using the generalized L-curve method as proposed in Murat *et al.* (2002). However, this method is computationally costly, and in practice the selection of multiple controlling parameters is mainly based on trial-and-error (e.g. Gallardo & Meju 2004) or simple criteria (e.g. Commer & Newman 2009).

In this study, we suggest a two-step method for the determination of the regularization parameter λ and data weighting parameter β. The method is illustrated in Fig. 3 using the synthetic example presented in Section 4. In this two-step method, the regularization parameter λ is first fixed, for example, as an intermediate value. Then, the joint inversion is carried out with β varying over a broad range, for example, from 10^{-3} to 10^3 in Fig. 3(a). By plotting the data misfit and β, a reasonable trade-off can then be chosen based on the shape of the curve. For instance, in Fig. 3(a) the optimal β is chosen to ensure the data misfit reaches a minimum. The second step is to determine the regularization parameter λ. Similarly, the joint inversion is conducted with the optimal β and a varying λ. The parameter λ and the related data misfit can be plotted to form a curve where the minimum can be chosen, for example, as shown in Fig. 3(b). We chose the associated λ as the optimal regularization parameter. Other methods such as the conventional L-curve method and its updated version (Hansen *et al.* 2007) can also be used here to determine the optimal regularization parameter.

4 A SYNTHETIC SAMPLE

In this section, a synthetic porous sample with a bimodal pore size distribution (see Fig. 4a) is used to demonstrate the applicability of the proposed joint inversion approach. The density of the pore size is modelled by a combination of two lognormal distributions. The related mean pore sizes are 20 μm and 2 μm; the standard deviations are 0.3 and 0.3; and the relative weighting is 0.4 and 0.6. Numerical experiments are conducted to produce the NMR responses [using eqs (1) and (4)] and SIP responses [using eqs (7) and (13)] of the sample. The NMR and SIP measurements are then jointly inverted to estimate the pore size distribution. Two cases were tested and are presented here. In Case 1, the numerical experiments and the inversion use the same petrophysical models. In Case 2, it is assumed we lack information on the petrophysical models. Thus, the inversion uses simplified petrophysical models, which are different from those used in the numerical experiments.

4.1 Case 1: inversion with exact petrophysical models

In Case 1, the NMR T_2 relaxation was numerically calculated using eqs (1) and (4) and the 'measured' decay curve is shown in Fig. 4(b). In the NMR experiment, the initial signal E_0 is set as 0.5 μV; the surface relaxivity ρ_s is set as 10 μm s^{-1}; the NMR signal was recorded for 5 s and the total number of the measurements are 3000. The random Gaussian errors were added to the simulated decay curve and the simulated noise level is 1 per cent. The complex conductivity of the sample is numerically calculated using eqs (7) and (13), and the 'measured' real and imaginary conductivities are shown in Figs 4(c) and (d), respectively. In SIP experiment, the following petrophysical parameters were used: $\sigma_0 = 0.045$ S m^{-1}, $\sigma_\infty = 0.05$ S m^{-1}, $D^+ = 1.32 \times 10^{-9}$ m^2 s^{-1}, $\varepsilon_{\mathrm{inf}} = 100\varepsilon_0$ (where ε_0 is the vacuum dielectric permittivity, 8.85×10^{-12} F m^{-1}). The

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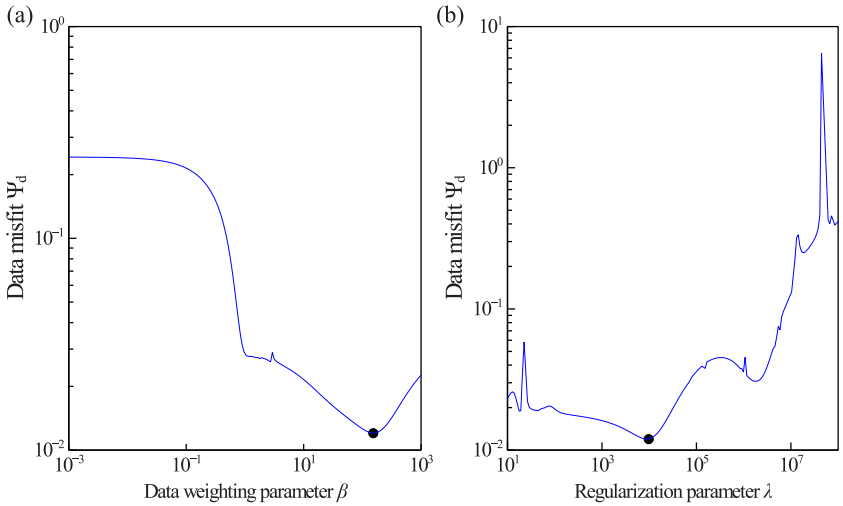


Figure 3. The determination of the data weighting parameter β and regularization parameter λ in the synthetic example presented in Section 4. The circle symbols indicate the positions of the optimal β and λ.

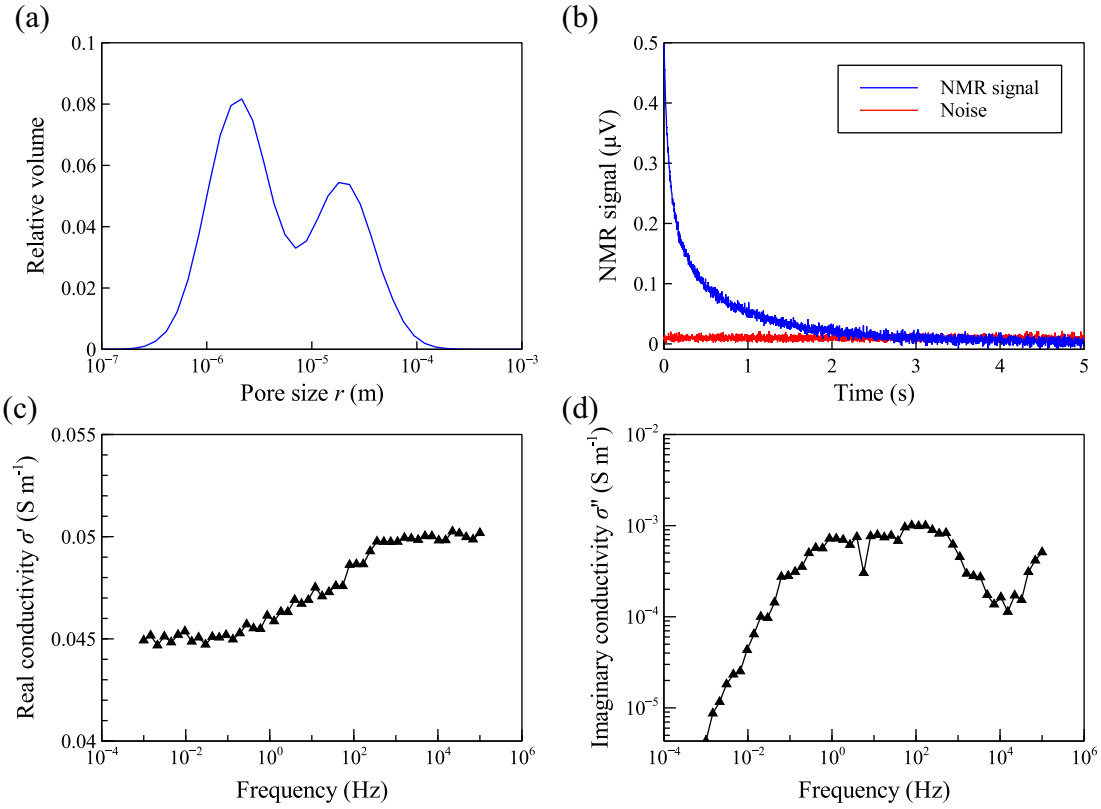


Figure 4. The synthetic example in Case 1: (a) the pore size distribution, (b) the 'measured' NMR T_2 relaxation, (c) the 'measured' real conductivity spectra and (d) the 'measured' imaginary conductivity spectra.

use of $100\varepsilon_0$ as the high frequency permittivity ε_{inf} is consistent with experimental measurements (e.g. see Jougnot *et al.* 2010; Kremer *et al.* 2016). The origin of this abnormally high value (for comparison, the permittivity of pure water is $\sim 79\varepsilon_0$) is still unclear and it could be caused by Maxwell-Wagner polarization, grain shape influence, inductive electromagnetic coupling or electrode influence (Kremer *et al.* 2016). The applied minimum and maximum frequencies are 10^{-3} Hz and 100 kHz, respectively, and the total number of SIP measurements is 50. Similarly, random Gaussian errors are added to the calculation to simulate the measurement noise. The noise level is 1 per cent for real conductivity and 35 per cent for imaginary conductivity. Three individual calculations were conducted for each measured frequency. The mean value was used as the measured conductivity and the standard deviation was used as the measurement error as shown in Figs 4(c) and (d).

The proposed joint inversion approach was used to estimate the pore size distribution of the sample based on the NMR and SIP measurements in Fig. 4. The proposed two-step method was used to determine the regularization parameter λ and data weighting parameter β. As shown in Figs 3(a) and (b), the reasonable trade-off

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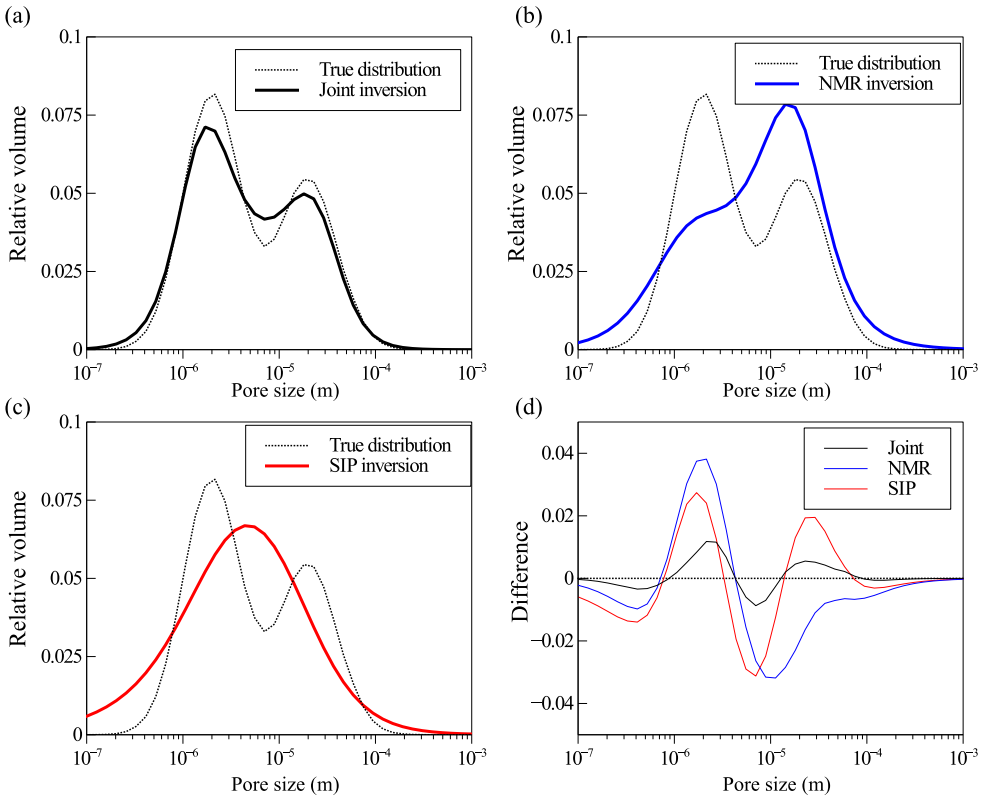


Figure 5. The inversion results of Case 1: (a) joint inversion, (b) separate NMR inversion, (c) separate SIP inversion and (d) discrepancies between inverted and true distributions of the pore size.

was chosen when the data misfit reaches a minimum, and the determined optimal λ and β are 9771 and 153, respectively. In the joint inversion, the surface relaxivity was taken as the true value, that is, 10 μm s^{-1}; and therefore, it was not updated during the inversion. The joint-inversion result is shown in Fig. 5(a). Separate NMR and SIP inversions were also carried out with the same regularization parameter, and the results are shown in Figs 5(b) and (c). The true density (normalized value) of the pore size is also plotted in Fig. 5 as well as discrepancies between the inverted and true densities.

As shown in Fig. 5(a), the joint inversion successfully recovers the bimodal shape of the pore size distribution. The estimated positions of the two peaks are very close to the true locations although the related densities are slightly smaller than the true values. In contrast, the results from separate inversions have very low resolutions. The NMR inversion, to a certain extent, recovers the bimodal nature of the pore size but it does not correctly locate the dominant pore size; the SIP inversion renders a pore size distribution with only one peak. Comparing with separate inversions, the joint inversion significantly improves the resolution of the estimated pore size distribution. Clearly, this improvement is due to the addition of extra measurements in the inversion.

We also tested whether the joint inversion can correctly estimate the surface relaxivity of the sample by allowing ρ_s to update during the inversion. The initial value of ρ_s was set as 1 μm $s^{-1}$and other parameters were unchanged. The inverted pore size distribution is very close to the one presented in Fig. 5(a). The evolution of ρ_s during the inversion is plotted in Fig. 6. It shows that although the initial ρ_s is relatively small, after several iterations, it approaches to the true value. The final inverted surface relaxivity is 9.47 μm s^{-1}, which is very close to the true value 10 μm s^{-1}.

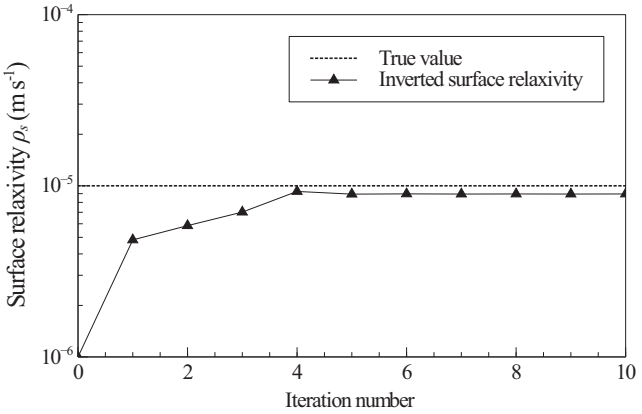


Figure 6. The surface relaxivity evolution during the joint inversion in Case 1.

4.2 Case 2: inversion with inaccurate petrophysical models

In Case 2, we use eqs (1) and (3) to simulate the NMR response of the sample so that the NMR relaxation is not necessarily within the fast diffusion regime. In SIP experiments, a Cole–Cole model is used to describe the unknown polarization that might be responsible for the anomalously high permittivity at intermediate frequencies (e.g. Lesmes & Morgan 2001; Kremer *et al.* 2016). Thus, the complex conductivity is calculated by combing eqs (7) and (13) and the added Cole–Cole model. The NMR and SIP responses calculated using these 'realistic' petrophysical models are shown in Fig. 7. In the calculation, same model parameters were used here except that the surface relaxivity is changed to 100 μm s^{-1}.

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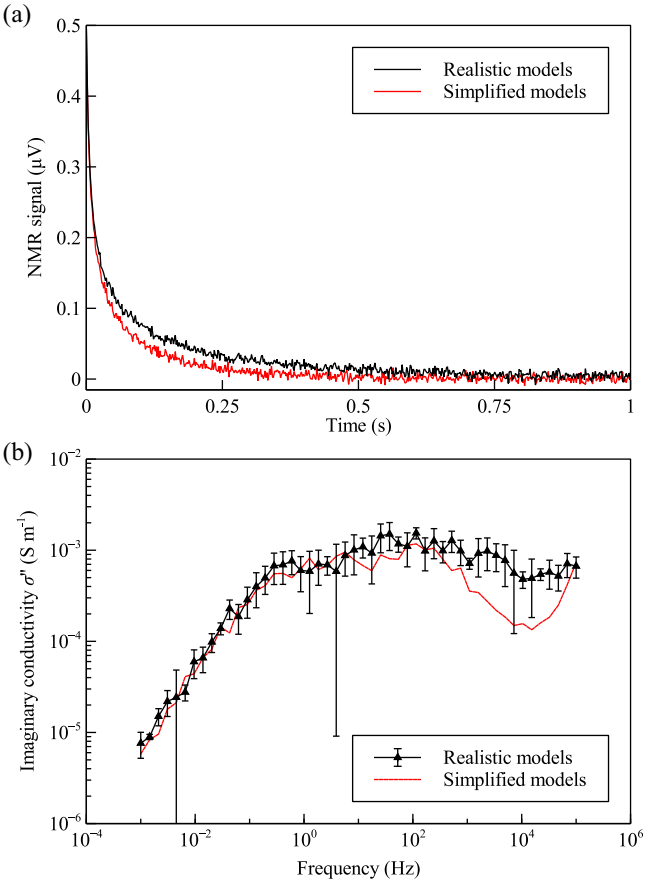


Figure 7. The measured (a) NMR and (b) SIP responses of the synthetic sample in Case 2. Compared with simplified petrophysical models, the realistic ones relax the fast diffusion condition and consider the unknown polarization responsible for anomalously high imaginary conductivity at intermediate frequencies.

For comparison, NMR and SIP responses calculated using the simplified petrophysical models (i.e. those used in Case 1) were also presented. As shown in Fig. 7, the 'realistic' and simplified petrophysical models give different NMR and SIP responses. The former relaxes the fast diffusion condition, and thus the related NMR relaxation rate is apparently slower (Fig. 7a). Moreover, 'realistic' petrophysical models give relatively large imaginary conductivities at frequencies larger 100 Hz (Fig. 7b), and the spectrum resembles experimental measurements shown in Lesmes & Morgan (2001).

The NMR and SIP responses (simulated from 'realistic' petrophysical models) are inverted to estimate the pore size distribution jointly or separately. In the inversions, it is assumed that we do not know the exact petrophysical relations and the simplified petrophysical models [eqs (1), (4), (7), and (13)] are used. The estimated pore size distributions from joint and separate inversions are shown in Fig. 8. The joint inversion still recovers the bimodal nature of the pore size and two peak locations are also well defined. However, the estimated densities are significantly lower than the true values. In addition, the inversion gives a very broad range for the pore size distribution. Compared to Fig. 5(a), the joint inversion results in Fig. 8(a) differ notably from the true model. This is attributed to the use of the simplified petrophysical models in the inversion. Again, we compare the results of joint inversion to those from separate inversions. It appears the separate inversion, neither SIP nor NMR inversion, can capture the bimodal nature of the pore size. The joint inversion still improves the quality of the estimated pore size distribution despite that we lack enough information on the exact petrophysical models of the sample.

5 APPLICATION TO BEREA SANDSTONE

In this section, we apply the proposed joint inversion approach to a Berea sandstone. The estimated pore size distribution is compared with those from separate inversions.

5.1 Sample and measurements

The well-characterized Berea sandstone is chosen in this study because both NMR and SIP data are available. The sample has a porosity ϕ of 20.2 per cent and the specific surface area S_m is determined with nitrogen adsorption as 0.5 $m^2\ g^{-1}$. Previous research shows quartz is the most abundant mineral in the sandstone and it also contains a significant amount of clays (e.g. Knight & Nur 1987). The intrinsic formation factor F of the sample is 16.3 as determined from multi-salinity measurement. The sample was saturated with NaCl solution ($\sigma_w = 0.043$ S m^{-1}) for NMR and SIP measurement. The NMR T_2 relaxation of the sample was measured using a 2 MHz Magritek Rock Core Analyzer^{TM} with CPMG pulse sequence and the echo spacing is 100 μs. The complex conductivity of the sample was measured in the frequency range between 10^{-3} Hz and 10 kHz using the SIP-LAB-II instrument (Radic). The NMR and SIP responses of the sample are shown in Fig. 9.

5.2 Inversion results

The NMR and SIP data of the sample in Fig. 9 are jointly inverted to estimate the pore size distribution of Berea sandstone. The data weighting and regularization parameters β and λ are determined using the proposed two-step method as shown in Fig. 10. In the inversion, the diffusion coefficient of ions in the EDL D^+ was taken as 3.2×10^{-12} $m^2\ s^{-1}$ as suggested by Revil (2012), considering that the sample has a significant amount of clays; the value may need to be adjusted by trial-and-error in cases where the calculated and measured peak frequencies in SIP spectra do not coincide. The surface relaxivity is allowed to change in the inversion and the initial value is set as $\rho_s = 5$ μm s^{-1}. However, we find ρ_s does not change during the inversion. The reason could be that the initial value was too close to the true surface relaxivity of the sample. The recovered petrophysical parameters from joint inversion are as follows: $\sigma_\infty = 0.0028$ S m^{-1}, $\sigma_0 = 0.0026$ S m^{-1}, $\varepsilon_{\text{inf}} = 128\varepsilon_0$, $E_0 = 7$ μV, $\rho_s = 4.995$ μm s^{-1}. For comparison, separate inversions are also conducted. In NMR inversion, the surface relaxivity ρ_s is kept as 5 μm s^{-1}; in SIP inversion, the diffusion coefficient D^+ is taken as 3.2×10^{-12} $m^2\ s^{-1}$. The regularization parameters used in separate inversions are individually determined using the traditional L-curve method so they may be different from that used in the joint inversion. The estimated pore size distributions from separate and joint inversions are shown in Fig. 11(a). The NMR and SIP responses calculated from the inversion results are also plotted in Fig. 9 for comparison.

The microstructure of Berea sandstone has been extensively studied using different methods, including MICP (Shi *et al.* 2011) and X-ray CT scan (Dong & Blunt 2009). Here, we use the sample in

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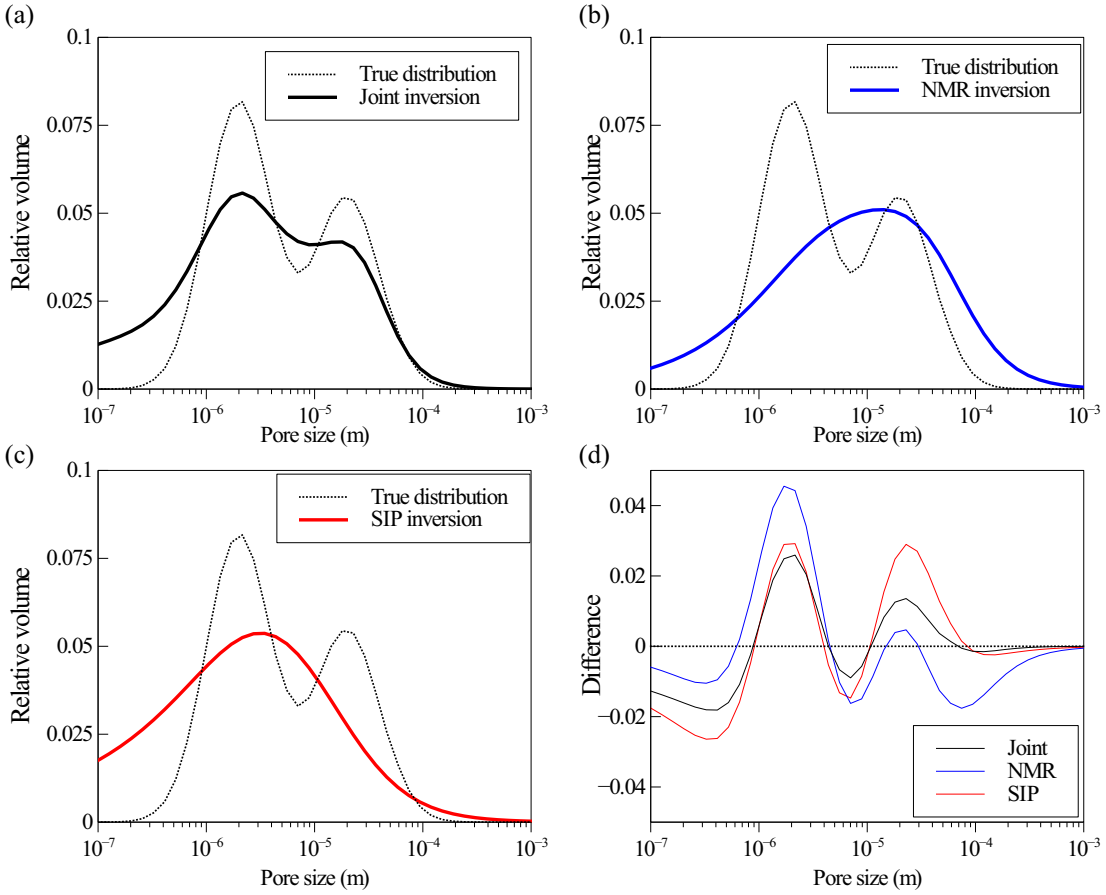


Figure 8. The inversion results of Case 2: (a) joint inversion, (b) separate NMR inversion, (c) separate SIP inversion and (d) discrepancies between inverted and true distributions of the pore size.

Shi *et al.* (2011) as an example to show the main features of pore (throat) size of Berea sandstone. The sample has a porosity of ∼19 per cent, which is very close to the sample in our study (∼20 per cent). The pore (throat) size distribution measured from MICP (Shi *et al.* 2011) is plotted in Fig. 11(b). One profound feature of the sample is the narrow range of the pore size, which centres on ∼10 μm. In contrast, the pore size distribution estimated from separate SIP inversion is quite broad, covering more than 2 orders. Comparing to MICP results, the SIP inversion significantly overestimates the contribution of small pores ($r < 3$ μm) although the imaginary conductivity spectrum is well reproduced base on this broad pore size distribution (Fig. 9b).

As shown in Fig. 11, the pore size estimated from joint inversion is consistent with the MICP results; both the shape and the relative position of the curve are well recovered, demonstrating the applicability of the joint inversion approach. The NMR response calculated from joint inversion fits the measurement perfectly. The simulated imaginary conductivity spectra agree with the measurements only at low (<0.1 Hz) and high frequencies (>10 kHz). At intermediate frequencies, the simulated σ'' is much smaller than the measurements. The discrepancy may be explained by the presence of other polarization, which is responsible for the measured quadrature conductivity at intermediate frequencies (Kremer *et al.* 2016). The estimated pore size distribution from NMR inversion is very similar to that of joint inversion, indicating that more weights are put on NMR data during the joint inversion. That implies, for this sample, the NMR data contain more information on the pore size than the SIP measurements.

5.3 Discussion

In this subsection, we briefly discuss limitations of the proposed joint inversion approach and possible directions for further studies. First, we point out that the Stern layer polarization model (i.e. eq. 5) may only be valid for relatively large pores (>∼0.1 μm) because it assumes the EDL thickness is much smaller than the radius of the curvature (see Schwarz 1962 or Appendix A in this study). Therefore, for geomaterials characterized by small pore size (<∼0.1 μm) such as mudstones or clayey soils, the Stern layer polarization model may not be able to describe the measured SIP spectrum. In that case, the joint inversion of SIP and NMR data with our approach may not improve the estimation of pore size distribution if compared with separate NMR porosimetry. In addition, since both NMR and SIP porosimetries assume isolated pores, the joint inversion approach may not work properly for geomaterials with well-connected heterogeneous pores such as loosely packed soils.

Despite the fact that the Stern layer polarization model has been frequently used in practice for various applications (e.g. Osterman *et al.* 2016; Leroy *et al.* 2017; Revil *et al.* 2017), it is still unclear whether it alone is sufficient to explain the measured complex conductivity of geomaterials over a broad frequency range. Thus, it is necessary to assess the applicability of the Stern layer polarization model for a given sample, and the proposed joint inversion approach is well suited for this purpose. Indeed, the simulated SIP response in Fig. 9(b) based on joint inversion of NMR and SIP data has indicated that Stern layer polarization may not explain the high imaginary conductivity of Berea sandstone at ∼100 Hz.

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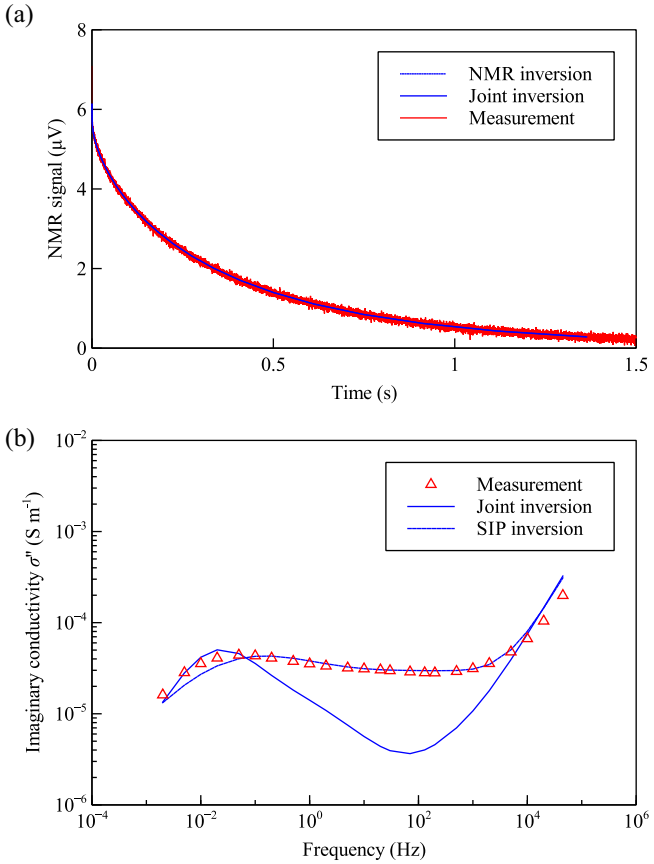


Figure 9. The measured and fitted (a) NMR and (b) SIP responses (imaginary conductivity) of the Berea sandstone sample. The fitted curves are from joint and separate inversions.

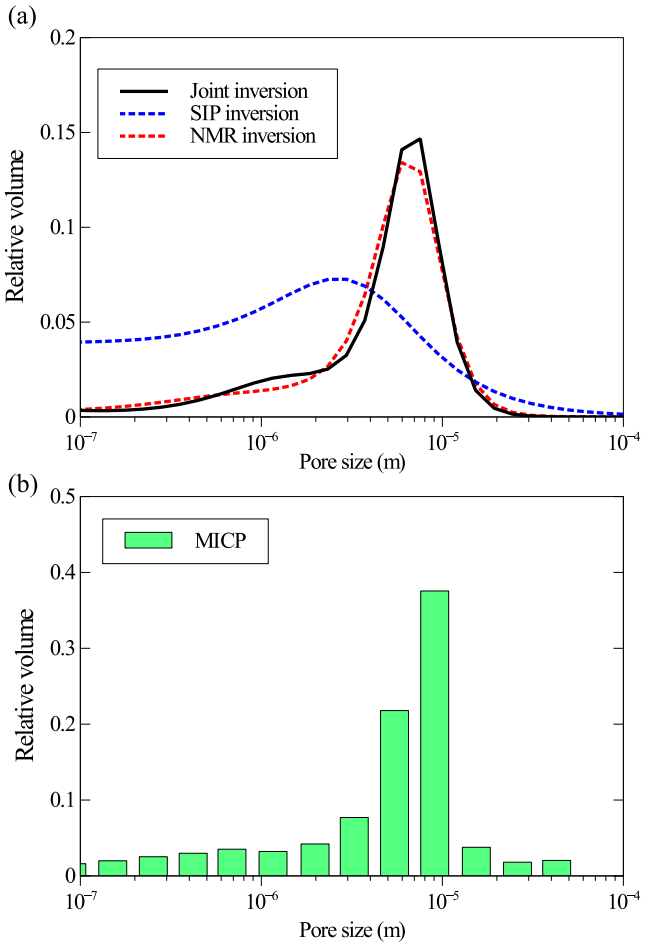


Figure 11. The estimated pore size distribution of the Berea sandstone using (a) SIP and NMR porosimetries and (b) mercury injection capillary pressure (MICP). The MICP-determined pore (throat) size distribution is from the Berea sandstone sample in Shi *et al.* (2011), which has a similar porosity ($\phi = 0.19$) with the one used in this study ($\phi = 0.2$).

In addition to EDL polarization, membrane polarization (Marshall & Madden 1959) is also commonly used to explain the induced polarization of porous media (e.g. Bairlein *et al*. 2016; Hördt *et al*. 2016). Different from Stern layer polarization, which relates the relaxation time to pore/particle size, membrane polarization suggests that the relaxation time may also be controlled by the geometry of the pore throat (Bücker & Hördt 2013). This conclusion was also supported by experimental evidence (e.g. Scott & Barker 2003). Considering that pore throat is usually smaller than pore space in geomaterials, membrane polarization could potentially explain the abnormal high imaginary conductivity at intermediate frequencies reported in Lesmes & Morgan (2001) and Kremer *et al.* (2016). In

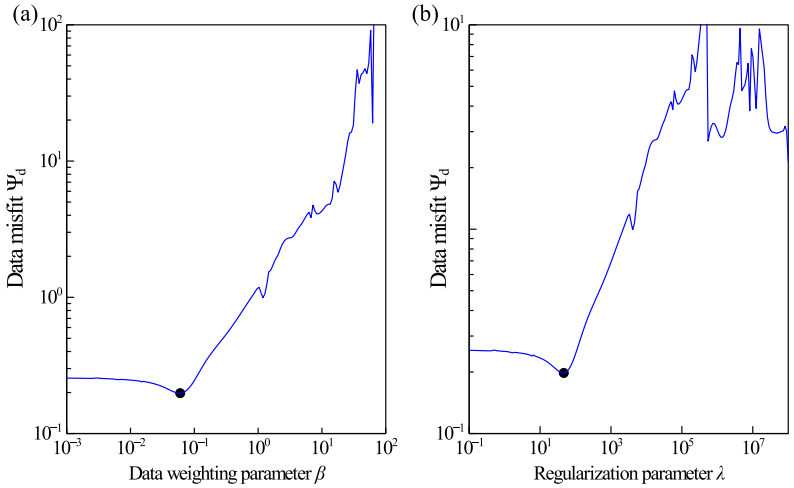


Figure 10. The determination of the data weighting parameter β and regularization parameter λ for the Berea sandstone sample. The circle symbols indicate the positions of the optimal β and λ.

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this perspective, the joint inversion incorporating membrane polarization would give additional information on the pore geometry of a geomaterial. Therefore, our current work might constitute a first step towards a comprehensive joint inversion that can extract the full pore geometry information from NMR and SIP data.

6 CONCLUSIONS

In this study, we propose to jointly invert NMR and SIP measurements for the estimation of the pore size distribution of geomaterials. The joint inversion is based on the fact that both NMR T_2 relaxation and low-frequency induced polarization of geomaterials are influenced by geometric characteristics of the pore space. In the joint inversion, the NMR T_2 relaxation time and the SIP relaxation time τ are linked through their petrophysical models. The Tikhonov regularization is used to constrain the smoothness of the pore size distribution. The trade-offs between two data sets and the balance between data misfit and model regularization are determined using a two-step method. The inversion problem is solved iteratively using the Gauss–Newton method and the related sensitivity matrix at each iteration is determined analytically.

Numerical examples have shown that the joint inversion could recover the main features of the pore size of the synthetic sample even if petrophysical models used in the inversion are inaccurate. Comparing to separate inversions, the addition of extra data set can improve the resolution of the estimated pore size distribution. In addition, the results from synthetic example indicate that the surface relaxivity could be properly estimated by the joint inversion when the diffusion coefficient of ions in the EDL is known. The proposed method has been applied to a Berea sandstone sample and the estimated pore size distribution is fairly consistent with that estimated from mercury injection capillary pressure measurements, demonstrating the applicability and usefulness of the proposed method.

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APPENDIX A: RELAXATION TIME OF EDL POLARIZATION

Consider a spherical surface Γ separating two domains Ω^i and Ω^e (Fig. A1). The charge density of the surface is Q_S, and the radius of the curvature of the surface is r_c. Apply an oscillating electric field $\mathbf{E}$ with its magnitude in the form of $E = E_0 e^{i\omega t}$ where ω is the angular frequency. Define spherical coordinates r and θ with $r = 0$ at the centre of domain Ω^i and $\theta = 0$ in the direction of $\mathbf{E}$.

The potential distribution in the domains φ^i and φ^e follows Laplace equation,

$$\nabla^2 \varphi^i = 0 \, (r < r_c) \tag{A1}$$

and

$$\nabla^2 \varphi^e = 0 \, (r > r_c). \tag{A2}$$

Define the potential at the surface Γ as φ_S. At least, eqs (A1) and (A2) satisfy the following boundary conditions:

(1) $\varphi^e \to -Er\cos\theta$ as $r \to \infty$,
(2) φ^i is finite as $r \to 0$, and
(3) $\varphi_S = \lim_{r=r_c} \varphi^e = \lim_{r=r_c} \varphi^i$.

The solutions to eqs (A1) and (A2) are well known and can be expressed as

$$\varphi^i = \sum \left(A_n^i r^n + B_n^i \frac{1}{r^{n+1}} \right) P_n(\cos\theta) \tag{A3}$$

and

$$\varphi^e = \sum \left(A_n^e r^n + B_n^e \frac{1}{r^{n+1}} \right) P_n(\cos\theta) \tag{A4}$$

where A_n^i, B_n^i, A_n^e, B_n^e are unknowns in the coefficients for the nth term of Legendre polynomials $P_n(\cos\theta)$. Applying boundary condition (1), eq. (A4) can be simplified as

$$\varphi^e = \left(A_1^e r + B_1^e \frac{1}{r^2} \right) \cos\theta. \tag{A5}$$

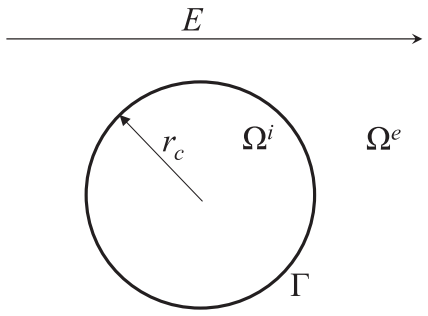


Figure A1. A spherical surface Γ separating two domains Ω^i and Ω^e. The radius of the curvature of the surface is r_c.

Then, the potential at the surface can be obtained as

$$\varphi_S = \left(A_1^e r_c + B_1^e \frac{1}{{r_c}^2} \right) \cos\theta. \tag{A6}$$

If we expand φ_S using Legendre polynomials, φ_S has the following form:

$$\varphi_S = \sum \beta_n P_n(\cos\theta), \tag{A7}$$

where β_n is the associated coefficient. Comparing eqs (A6) and (A7), we find that only the first term ($n = 1$) in eq. (A7) is non-zero. This result will be used later to determine the relaxation time of the change in Q_S in the surface Γ.

Assume the charge on Γ (a thin EDL) only moves tangentially. The movement of charge on Γ is due to two driving forces: (1) electric field and (2) ion concentration gradient. Considering the continuity, the resulting time derivative of the surface charge density can be written as

$$\frac{\partial Q_S}{\partial t} = e_0 \, \nabla \cdot (j_E + j_D), \tag{A8}$$

where e_0 is the charge carried by an ion in the EDL, and j_E and j_D are surface fluxes of ions induced by electric field and ion diffusion, respectively. Consider the change in Q_S is ΔQ_S, which is much less than Q_S. Also, consider the expressions of j_E and j_D. Then, eq. (A8) can be expressed as

$$i\omega \, \Delta Q_S = \frac{\mu k T}{{r_c}^2} \frac{1}{\sin\theta} \frac{\mathrm{d}}{\mathrm{d}\theta} \left(\sin\theta \frac{\mathrm{d}\Delta Q_S}{\mathrm{d}\theta} + \frac{e_0 Q_S}{kT} \sin\theta \frac{\mathrm{d}\varphi_S}{\mathrm{d}\theta} \right), \tag{A9}$$

where μ is the mobility of ions in EDL, k is Boltzmann constant and T is the absolute temperature. We also expand the surface charge variation ΔQ_S using Legendre polynomials as

$$\Delta Q_S = \sum \alpha_n P_n(\cos\theta). \tag{A10}$$

Insert eqs (A7) and (A10) into eq. (A9), and after some derivative calculations, eq. (A9) becomes

$$i\omega \sum \alpha_n P_n = \frac{\mu k T}{{r_c}^2} \sum \alpha_n \left(\frac{\cos\theta}{\sin\theta} P_n' + P_n'' \right) + \frac{\mu e_0 Q_S}{{r_c}^2} \sum \beta_n \left(\frac{\cos\theta}{\sin\theta} P_n' + P_n'' \right). \tag{A11}$$

Considering the following equality,

$$P_n'' + \frac{\cos\theta}{\sin\theta} P_n' + n(n+1) \, P_n = 0, \tag{A12}$$

eq. (A11) can be rewritten as

$$\sum \alpha_n \left(i\omega + \frac{\mu k T}{{r_c}^2} n(n+1) \right) P_n = \sum \beta_n \frac{\mu e_0 Q_S}{{r_c}^2} n(n+1) P_n. \tag{A13}$$

Because of the orthogonality of Legendre polynomials, the following relation is valid for all the terms,

$$\alpha_n = -\frac{1}{1 + i\omega \frac{{r_c}^2}{n(n+1)\mu k T}} \frac{e_0 Q_S}{kT} \beta_n. \tag{A14}$$

As discussed, the coefficient β_n for the surface potential φ_S is non-zero only when $n = 1$, we have the following relation between ΔQ_S and φ_S:

$$\Delta Q_S = -\frac{1}{1 + i\omega\tau} \frac{e_0 Q_S}{kT} \varphi_S \tag{A15}$$

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with

$$\tau = \frac{{r_c}^2}{2\mu kT} = \frac{{r_c}^2}{2D^+} \tag{A16}$$

where D^+ is the diffusion coefficient of the ions in the EDL.

Eq. (A16) states that the surface charge density follows a simple Debye relaxation with relaxation time τ, which is controlled by the radius of the curvature of the surface Γ. In the derivation, the electrical properties of domains Ω^i and Ω^e were not specified, and therefore, eq. (A15) holds for both following cases: (1) domain Ω^i is solid particle and domain Ω^e is electrolyte; and (2) domain Ω^i is electrolyte and domain Ω^e is solid particle. In the former case, the radius of the curvature of the surface Γ is equal to the particle radius; for the latter, r_c is equal to the pore size.

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文献 3/4

Physical Explanation of Archie's Porosity Exponent in Granular Materials: A Process-Based, Pore-Scale Numerical Study

Qifei Niu; Chi Zhang

Geophysical Research Letters (2018); DOI: 10.1002/2017GL076751

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Key Points:
- Pore-scale modeling reveals that Archie's porosity exponent in dilute granular materials is exclusively related to particle properties
- For nondilute granular materials, the exponent *m* increases continuously as the porosity decreases
- Numerical results suggest a unique correlation between pore throat volume and normalized *m* in nondilute granular materials

Supporting Information:
- Supporting Information S1

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Physical Explanation of Archie's Porosity Exponent in Granular Materials: A Process-Based, Pore-Scale Numerical Study

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Abstract The empirical Archie's law has been widely used in geosciences and engineering to explain the measured electrical resistivity of many geological materials, but its physical basis has not been fully understood yet. In this study, we use a pore-scale numerical approach combining discrete element-finite difference methods to study Archie's porosity exponent *m* of granular materials over a wide porosity range. Numerical results reveal that at dilute states (e.g., porosity $\phi >$ ~65%), *m* is exclusively related to the particle shape and orientation. As the porosity decreases, the electric flow in pore space concentrates progressively near particle contacts and *m* increases continuously in response to the intensified nonuniformity of the local electrical field. It is also found that the increase in *m* is universally correlated with the volume fraction of pore throats for all the samples regardless of their particle shapes, particle size range, and porosities.

Plain Language Summary In many topical areas in Earth science and engineering such as groundwater hydrology and petroleum explorations, the electrical resistivity data are widely used to estimate the porosity and/or water/oil content in the subsurface with an empirical relationship that originated in the 1940s of the last century. This relation, known as Archie's law, relates a material's resistivity at saturation to its porosity and the pore fluid resistivity with a power function. As an empirical parameter, the exponent of the function varies considerably among different geological materials, and therefore, the estimated porosity/fluid content is significantly affected by the chosen value. At present, a physical understanding of the exponent is still limited mainly due to the lack of the detailed information on the complex geometry of the pore space. Here we use computational simulations to obtain the microscale geometrical characteristics of synthetic granular materials and to analyze the dominant parameter(s) impacting the exponent. We also provide a pore-scale, mechanistic explanation for the changes in the exponent of granular materials subjected to geological compaction. The findings from this study would help develop a physics-based approach to determine the value of the exponent in the modeling of the resistivity of various Earth materials in practice.

1. Introduction

Electrical and electromagnetic methods play an essential role in imaging and monitoring various properties and processes in both deep and shallow subsurface of the Earth (e.g., Key et al., 2013; St. Clair et al., 2015). To link the measured resistivity to other properties of interest, considerable efforts have been made to develop constitutive models for geological materials (e.g., Bussian, 1983; Ghanbarian et al., 2014; Glover et al, 2000; Sen et al., 1981). The most widely used model is probably Archie's law (Archie, 1942), which relates the conductivity σ of rock to the porosity ϕ by

$$\sigma = \frac{\sigma_w}{F} = \sigma_w \phi^m \tag{1}$$

where σ_w is the pore fluid conductivity, *F* (i.e., ratio of σ_w to σ) is the formation factor, and *m* is the porosity exponent. Archie (1942) found that typical value of *m* is between 1.8 and 2 for sandstones and ~1.3 for unconsolidated sand; however, there are many observations of *m* deviating from the above values (e.g., Friedman, 2005). Equation (1) assumes that rock grains are insulators, and it only considers the current flowing through the bulk solution. For geomaterials with charged surface, the contribution of surface conduction (Revil & Glover, 1998) needs to be excluded from σ to calculate the "intrinsic" value of *F* (Waxman & Smits, 1968).

The physical basis for Archie's law has been the subject of many studies (Hunt, 2004; Mendelson & Cohen, 1982; Sen et al., 1981; Sheng, 1990; Wong et al., 1984), and one key scientific challenge is to reveal the dominant textural control on *m* (Glover, 2009). Theoretical studies (e.g., Mendelson & Cohen, 1982; Sen et al., 1981)

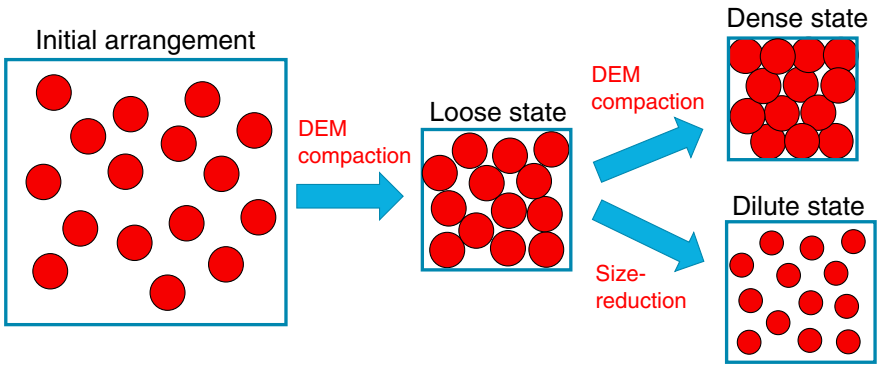


Figure 1. Schematic diagram illustrating the procedure to prepare various states of the samples.

have confirmed the strong dependence of particle properties on *m* as observed in experiments (e.g., Jackson et al., 1978). In addition, numerical and experimental studies have shown profound influences of pore attributes on *m*, including porosity (Borai, 1987; Olsen et al., 2008), tortuosity (Salem & Chilingarian, 1999), the skewness of the pore size distribution (Wong et al., 1984), and connectivity (Bernabé et al., 2011). Moreover, it has also been suggested that *m* might be related to some geological processes such as cementation (Archie, 1942; Leroy et al., 2017; Schwartz & Kimminau, 1987; Wu et al., 2010), compaction (Fatt, 1957; Glover et al., 2000), and mineral dissolution (Garing et al., 2014; Sen et al., 1997). Indeed, the above mentioned factors are not independent but are essentially interrelated in affecting *m*. At present, there is no consensus on the physical meaning of *m* in geological materials and further studies are still necessary to reconcile the existing findings.

In this study, we investigate the controls on *m* in granular materials using a pore-scale numerical approach that combines discrete element-finite difference methods. The samples presented here consist of spherical and oblate particles, and the solid-phase concentration varies broadly, from dilute (i.e., a limited particle-particle interaction; $\phi >$ ~65%) to dense states (i.e., $\phi =$ ~30%). Our work is unique in applying a physics-based approach, that is, discrete element method (Cundall & Strack, 1979), to simulate the compaction of loosely packed granular samples. The effective conductivity of the granular sample is calculated by directly solving the finite difference representation of the Laplace equation (Garboczi, 1998). Furthermore, we use a modified maximal ball algorithm (Dong & Blunt, 2009) to determine the pore attributes of the samples such that the possible link between material's texture and *m* can be revealed.

2. Samples and Methods

2.1. Sample Preparations

Six granular samples are used in the 3-D numerical simulation, and they are assemblies of 8,000 spherical or oblate particles. In each sample, all the particles have the same shape ($e = 1$, 0.67, or 0.5 where e is the ratio of the semiminor to semimajor axes), and the particle size follows either a Dirac delta distribution centered at d or a uniform distribution in the range $[0.5d, 1.5d]$. For all the samples, we use the same procedure to attain a number of states as shown in Figure 1. We start with an initially sparse arrangement, in which the particle locations and orientations are random. The loose state of the sample (e.g., $\phi =$ ~42%) is then obtained by isotropically reducing the boundary dimensions until the sample develops a loading-bearing fabric (e.g., see Kuhn et al., 2014; Sun et al., 2013). Afterward, the dense state of the sample ($\phi =$ ~30%) is attained by further compressing the loose granular assembly. The discrete element method code OVAL developed by Kuhn (2006) (available at http://faculty.up.edu/kuhn/oval/oval.html) is used to model this process, and the details of the simulation are further explained in the supporting information. In addition, we also attain the dilute states (e.g., $\phi >$ ~65%) of the samples by simultaneously reducing the size of all the particles but fixing their locations and orientations at loose states (Figure 1).

2.2. Effective Conductivity Calculation

To calculate the effective electrical conductivity of a sample, a cubic representative elementary volume (REV) is taken out from the granular assembly. The REV is then discretized into a number of voxels with the same

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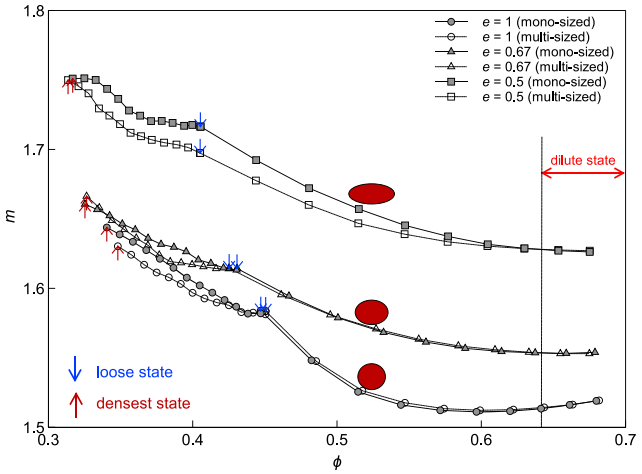


Figure 2. The variation of Archie's porosity exponent m of six granular samples from dilute to dense states.

dimension, and each voxel has a constant conductivity. Subjected to a uniform external electrical field, the spatial current density in the REV can be calculated by solving the finite-difference representation of Laplace equation. In the calculation, the grain conductivity is set to zero to comply with the assumption of equation (1). Based on the volume-averaged current density and electrical field, the effective conductivity of the sample can then be determined by Ohm's law. In this study, we adopted an open source code (DC3D.F) developed at the National Institute of Standards and Technology (available online at http://ciks.cbt.nist.gov/garbocz/manual/node52.html) to realize the numerical calculation (e.g., Andrä et al., 2013; Garboczi, 1998; Niu et al., 2017; Niu & Zhang, 2017; Zhan et al., 2010).

3. Results

The effective electrical conductivity σ of the six saturated granular samples is calculated at 22 states ranging from dilute to dense. In the calculation, the REV has a dimension of 3 mm × 3 mm × 3 mm, and it is discretized into 300^3 voxels. The related m of the samples is determined as $m = -\log_{\phi}(\sigma_w/\sigma)$, and the results are presented in Figure 2. It appears that the exponent m of a granular material is not a constant but a function of ϕ. At dilute state, m roughly keeps constant; as ϕ decreases, m increases continuously. In addition, it is apparent that both particle shape and size range have strong effects on m at a given porosity. In the following, these features will be discussed in detail.

3.1. Effect of Particle Shape

The effect of particle properties on m has been well studied with the differential effective medium theory (Sheng, 1990), which implements a homogenization process that incrementally adds dilute particles to the liquid phase toward a given solid-phase concentration. For granular materials consisting of spheroids, Mendelson and Cohen (1982) have developed an analytical equation linking m to the particle shape and orientation. If the spheroids are randomly orientated, m can be written as (Mendelson & Cohen, 1982; Sen, 1984)

$$m = \frac{5 - 3L}{3(1 - L^2)} \tag{2}$$

where L is the depolarization factor along the semiminor axis, which is directly related to the particle shape (see supporting information). For particles aligned along an electric field with direction i, the related m_i is expressed as (Sen et al., 1981)

$$m_i = \frac{1}{1 - L_i} \tag{3}$$

where L_i is the depolarization factor along direction i. Note that the differential effective medium theory iteratively applies the dilute assumption, and therefore, in principle, equation (2) should only be valid for granular samples at dilute states.

Equations (2) and (3) are used to show the influence of particle shape on m as plotted in Figure 3. Note that equation (3) is for aligned particles, and therefore, the results in directions parallel (||) and perpendicular (⊥) to the semi-major axis can be regarded as two limiting cases of m. For comparison, the porosity exponent of the three monosized samples in Figure 2 at dilute states m_0 is also plotted in the figure. In addition, we include the results of other 29 synthesized dilute granular samples. These samples consist of randomly packed, monosized, oblate particles with e ranging between 0.3 and 1, and they are also prepared with the procedure in Figure 1. As shown in Figure 3, m_0 from

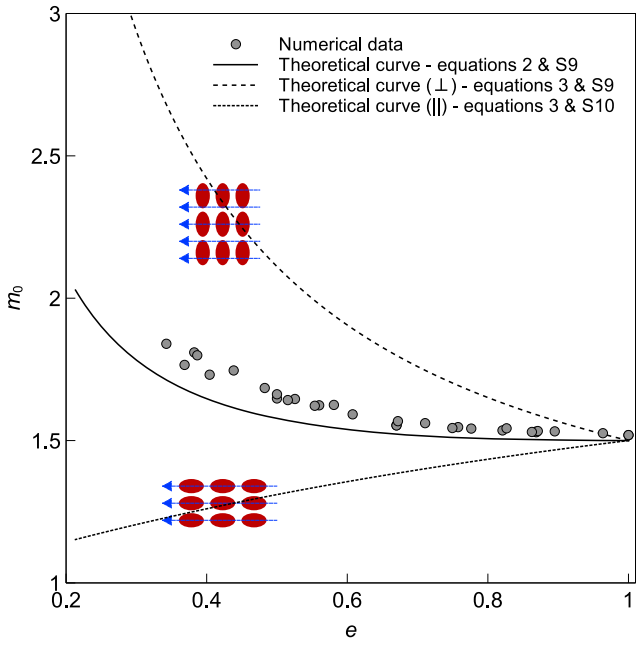


Figure 3. Influences of particle shape and orientation on Archie's porosity exponent of granular materials at dilute states m_0.

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numerical simulation and equation (2) shows a similar trend as e varies. In addition, m_0 from simulations is within the upper and lower boundaries described by equation (3). Both numerical and theoretical results indicate that the particle shape controls m_0. For granular materials with randomly oriented particles, as the particle becomes more aspherical, the related m_0 is larger. In general, numerically determined m_0 is slightly higher than theoretical values as shown in Figure 3. This minor discrepancy is due to the use of discrete voxels to represent continuum objects (Arns et al., 2002) in calculating σ (see more discussions in the supporting information).

3.2. Effect of Particle Size Range

Previous studies on the effect of particle size range on m are controversial. For instance, the effective medium theory suggests that m is independent of particle size range (Mendelson & Cohen, 1982; Sen et al., 1981). However, some experimental results have indicated that the particle size range can affect m (e.g., Lemaitre et al., 1988). We believe that the contrasting outcomes are due to different sample states considered. As noted, the differential effective medium theory iteratively applies the dilute assumption; therefore, in principle, their prediction is only applicable for dilute samples. In contrast, Lemaitre et al.'s (1988) experiments were using granular packings, and hence, the observed influence of particle size range on m should only be associated with nondilute samples.

The numerical results in Figure 2 are used to test our above hypothesis. At dilute state, we found that m of multisized and monosized samples with same particle shape is almost identical, suggesting limited effects of particle size range on m. This finding agrees with the differential effective medium theory. As the particles become more concentrated (i.e., porosity decreases), multisized and monosized samples start to have varying m values despite the same particle shape. It appears that samples with a broader particle size range (empty symbols in Figure 2) tend to have a lower m. This observed dependence of particle size range on m of nondilute granular samples is consistent with some experimental data, for example, as reviewed by Friedman (2005).

3.3. Evolution of *m*

In Figure 2, it is shown that the exponent m of granular materials increases with the solid-phase concentration but its physical causes have not been explained. Here we examine the m variation from a pore-scale perspective. Consider a cubic porous medium with length l subjected to an electric potential gradient. The formation factor F is defined as (Johnson et al., 1986)

$$F = \frac{l|\psi|^2}{\int|\mathbf{e}(\mathbf{r})|^2 dV} \tag{4}$$

where ψ is the potential drop across the sample and $\mathbf{e}(\mathbf{r})$ is the resulting electrical field in the pore space V. Considering equation (1), the microscopic definition of m can be expressed as

$$m = \frac{\ln\left(\int|\mathbf{E}(\mathbf{r})|^2 df\right)}{\ln(\int df)} \tag{5}$$

where $\mathbf{E}(\mathbf{r})$ is the local electric field $\mathbf{e}(\mathbf{r})$ normalized by ψ/l and df indicates that the integration is over the pore space from 0 to ϕ. According to equation (5), if the local electrical field in the pore space is uniform (e.g., in straight capillary tubes), m is unity because $|\mathbf{E}(\mathbf{r})| = 1$. For porous media with complex pore space, the electrical field is not uniform, which makes m deviate from 1. From this perspective, a change in m of a porous medium is therefore an indication of the spatial variations in $|\mathbf{E}|^2$ in its pore space.

Take the monosized sample with $e = 0.67$ as an example. The electric field in its pore space is calculated at five different states: A, B, C, D, and E. As shown in Figure 4a, A represents the dilute state ($\phi = 67\%$), C corresponds to the loose state ($\phi = 43\%$), E is the densest state ($\phi = 32\%$) after compaction, and B ($\phi = 50\%$) and D ($\phi = 39\%$) are two intermediate states. In Figure 4a, the histograms of $|\mathbf{E}|^2$ at all voxels in the pore space are plotted. It is shown that as the solid-phase concentration increases, the $|\mathbf{E}|^2$ distribution becomes more heterogeneous (or dispersed) and more pore space acquires a high $|\mathbf{E}|^2$ value. For instance, at state A, only a few voxels have a $|\mathbf{E}|^2$ value higher than 100 but at state E, the number increases to $\sim 2 \times 10^4$. The intensified nonuniformity of $|\mathbf{E}|^2$ distribution, according to equation (5), makes m continuously increase as the sample changes from states A to E as shown in Figure 2.

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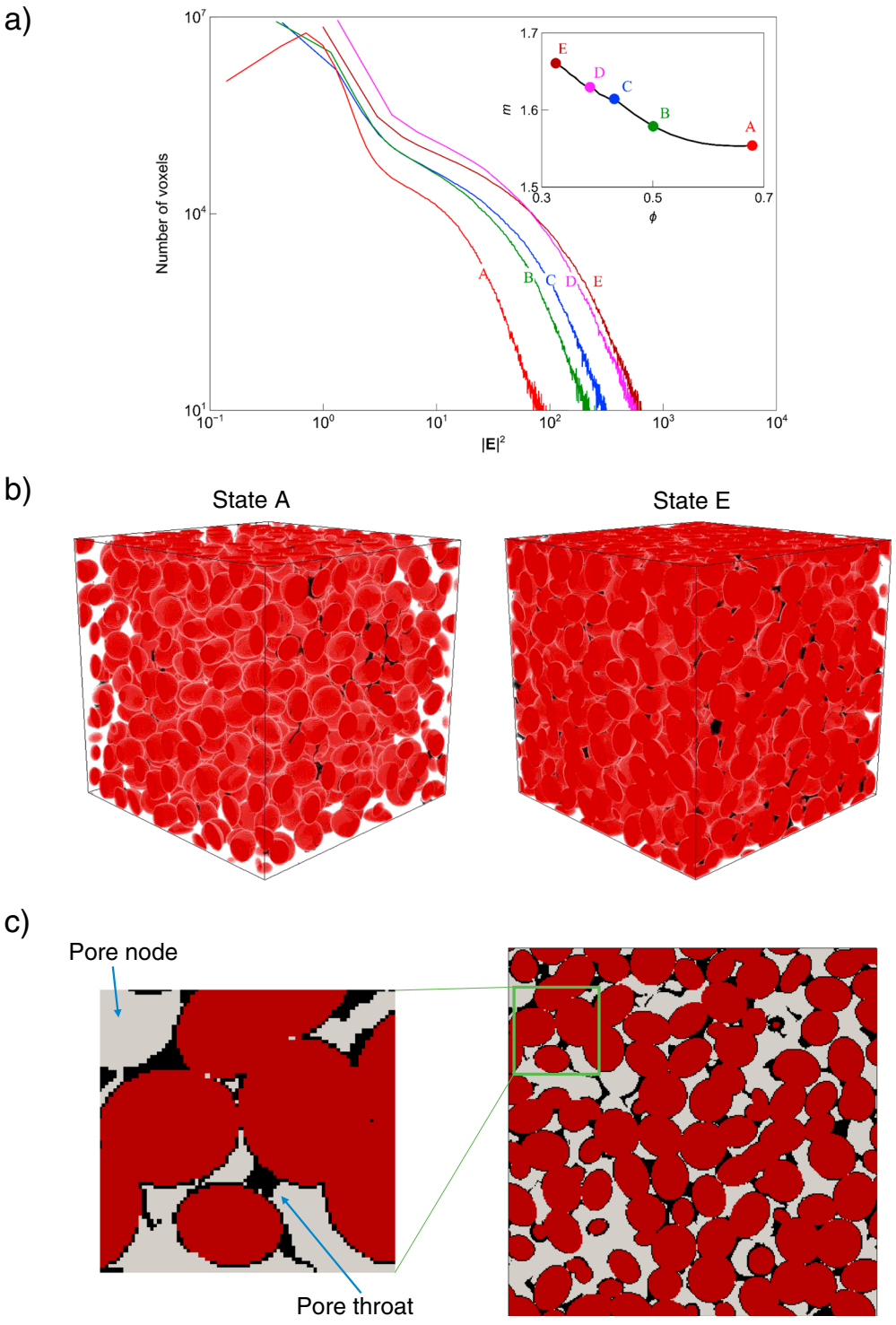


Figure 4. Pore-scale characteristics of the monosized sample with $e = 0.67$: (a) the number of voxels with a given $|\mathbf{E}|^2$ value at five states: A ($\phi = 67\%$), B ($\phi = 50\%$), C ($\phi = 43\%$), D ($\phi = 39\%$), and E ($\phi = 32\%$); (b) 3-D illustration of the microstructure and electric field at states A and E; and (c) 2-D illustration of the microstructure and electric field at state E.

To visualize, we plot the 3-D microstructure and spatial $|\mathbf{E}|^2$ distribution of the sample at states A and E in Figure 4b. In the images, solid particles are highlighted in red and pore space with $|\mathbf{E}|^2 > 50$ is featured in black. Comparing to state A, it is apparent that the sample at state E has a much larger portion of pore space with high $|\mathbf{E}|^2$, indicating intensified nonuniformity of the electrical field strength. Interestingly, we found that

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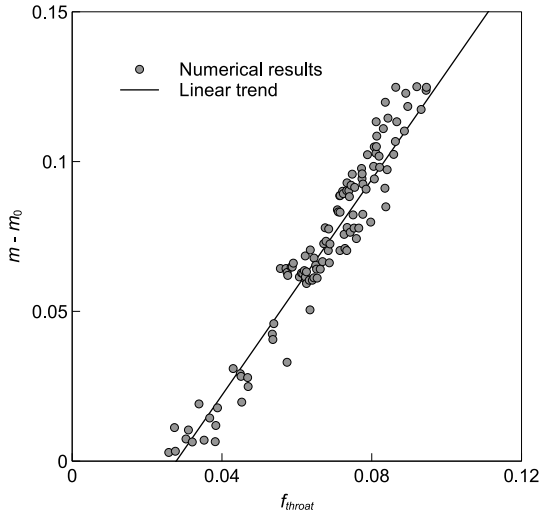


Figure 5. Correlation between the normalized Archie's porosity exponent m-m_0 and the volume fraction of pore throats f_{throat} for all the samples considered in this study.

most of the pores in black are within the region where two particles contact (i.e., the narrow pore throats as shown in Figure 4c). Since a large $|\mathbf{E}|$ is equivalent to a high electric flux density, Figure 4 elucidates that when the granular material moves from the dilute state to a dense state, the electrical flow becomes more concentrated in the pore throats.

4. Discussion

As discussed in section 3.3, the increase of m in granular materials is related to the intensified nonuniformity of the electric field strength $|\mathbf{E}|$ in the pore space and the concentrated electric flow at pore throats. This intrigues us to investigate if m and pore throat volume are correlated. We used a modified maximal ball algorithm (Dong & Blunt, 2009) to determine the pore throat volume for the granular samples. This algorithm applies a clustering process (Dong & Blunt, 2009) to organize the maximal balls into pore throat chains such that the profile of the pore space can be described. According to these chains, the pore space can be segmented into pore node and throats (e.g., see, Blunt et al., 2013).

The results are shown in Figure 5 where the volume fraction of pore throats f_{throat} and normalized porosity exponent m-m_0 are plotted for all the six samples at different states. We suggest to normalize m by subtracting the value at the dilute state m_0 such that individual effect of particles can be isolated from the impact of pore attributes. This treatment may also permit a unified explanation that reconciles that different interpretations of m already existed in the literature. As shown in Figure 5, all the samples collapse into a single linear line regardless of their particle shapes, size ranges, and porosities. This unique trend suggests that there might be a universal correlation between the increase in m (relative to the dilute state of individual samples) and the relative volume of pore throats for granular materials. Further studies are suggested to experimentally verify this correlation.

The positive relation observed in granular materials (Figure 5) can explain many experimental observations regarding m in consolidated sediments. For example, Wong et al. (1984) observed that fused glass beads sample with a broader pore size distribution tends to have a higher m value. From the micrographs in Wong et al. (1984), it is clear that the sample with a broad pore size range is also related to increased pore throat volume fraction. Another example is the Fontainebleau sandstone reported by Revil et al., (2014) where m generally becomes larger as the degree of cementation (silica overgrowth) increases (e.g., their Figure 5a). Their microcomputed tomography images clearly show that as the porosity decreases, the pore space becomes more heterogeneous (i.e., the fraction of pore throat volume increases). In this sense, it seems reasonable to extend our findings in granular materials to sedimentary rocks; however, further studies are still needed to quantitatively verify its applicability.

While applying Archie's law to interpret field resistivity data, m is usually chosen empirically with little theoretical justification. A better understanding of the physical significance of m will directly help geophysicists determine the appropriate values for m and F. In hydrocarbon explorations, an accurate m can significantly improve the reserve calculation from well-logging data (e.g., Glover, 2009; Shankar & Riedel, 2011). In hydrogeophysics, a correct F value is the key to characterizing permeability with either the characteristic pore size-based model (Johnson et al., 1986) or the Kozeny-Carman equation incorporating tortuosity (e.g., Slater et al., 2014). In solid Earth studies, the estimation of porosity and fluid budget in subduction zone from resistivity data will also benefit significantly from a physical understanding of m (Naif et al., 2016).

5. Conclusions

Based on the pore-scale, numerical study, we have reached the following conclusions regarding Archie's porosity exponent m of granular materials. At dilute states (e.g., $\phi > 65\%$), m is exclusively related to particle shape and orientation; this relationship can be quantitatively described with the differential effective

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medium theory. As the granular materials deviate from the dilute state toward a denser state, m increases with the solid-phase concentration; in this process, the particle size range has a noticeable effect on m, and samples with a wider size range tend to have a lower m at a given porosity. From dilute to dense states, the increase in m of granular materials is related to the pore-scale variations in the electrical field $|\mathbf{E}|$; in general, a high m value corresponds to a more dispersed distribution of $|\mathbf{E}|^2$ in the pore space. Our numerical results also suggest that for granular materials, a universal relation might exist, which links the volume fraction of pore throats to the increase in m (relative to the value at dilute state m_0) regardless of the particle shape, particle size range, and porosities.

Acknowledgments
The discrete element method code OVAL and finite difference code DC3D.F can be obtained online using the URLs listed in the main text. Other data can be obtained from the corresponding author upon request. The authors would like to thank Martin Blunt at the Imperial College London for providing the pore network extraction code.

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文献 4/4

Permeability Prediction in Rocks Experiencing Mineral Precipitation and Dissolution: A Numerical Study

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Permeability Prediction in Rocks Experiencing Mineral Precipitation and Dissolution: A Numerical Study

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Key Points:
- Fluid and electric flows are numerically simulated at the pore scale in rocks undergoing precipitation and dissolution
- Simulations show that rock permeability k can be fairly well estimated with the electrical tortuosity and an appropriate length scale
- The uncertainty of the k prediction is mainly from the electrical tortuosity, which can underestimate the true hydraulic tortuosity

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Abstract In this study, we focus on the electrical tortuosity-based permeability model $k = r_{eff}^2/8F$ (r_{eff} is an effective pore size, and F is the formation factor) and analyze its applicability to rocks experiencing mineral precipitation and dissolution. Two limiting cases of advection-dominated water-rock reactions are simulated, that is, the reaction-limited and transport-limited cases. At the pore scale, the two precipitation/dissolution patterns are simulated with a geometrical model and a phenomenological model. The fluid and electric flows in the rocks are simulated by directly solving the linear Stokes equation and Laplace equation on the representative elementary volume of the samples. The numerical results show that evolutions of k and F differ significantly in the two limiting cases. In general, the reaction-limited precipitation/dissolution would result in a smooth variation of k and F, which can be roughly modeled with a power function of porosity ϕ with a constant exponent. In contrast, the transport-limited precipitation/dissolution mostly occurs near the pore throats where the fluid velocity is high. This induces a sharp change in k and F despite a minor variation in ϕ. The commonly used power laws with constant exponents are not able to describe such variations. The results also reveal that the electrical tortuosity-based permeability prediction generally works well for rocks experiencing precipitation/dissolution if r_{eff} can be appropriately estimated, for example, with the electrical field normalized pore size Λ. The associated prediction errors are mainly due to the use of electrical tortuosity, which might be considerably larger than the true hydraulic tortuosity.

1. Introduction

In hydrology and many water-related disciplines, the distribution of permeability k in the subsurface is crucial for conducting a flow analysis, for example, to assess the aquifer performance under different water usage scenarios (Paniconi & Putti, 2015), to predict the organic solute plume front in contaminated sites (e.g., Rizzo & de Barros, 2017), and to evaluate different waterflooding strategies to recovery additional oils from reservoirs (e.g., Friesen et al., 2017). The permeability of a rock is primarily controlled by its texture (e.g., porosity, particle/pore size, sorting, and surface area) and mineralogy, and it can vary over several orders of magnitude (e.g., Brace, 1980) owing to the heterogeneous nature of the rock (Honarpour et al., 1995). In the last several decades, there has been a continuous effort to explore the links between k and the rock's texture (and/or mineralogy; e.g., Bryant & Blunt, 1992; Garboczi, 1990; Kwon et al., 2004; Mualem, 1976; Xu & Yu, 2008). These studies have significantly improved our capability of predicting a rock's permeability to understand the fluid flow phenomenon in the subsurface.

From a geological viewpoint, the permeability of a rock is not stationary but continuously evolves during diagenesis. This is because diagenetic processes, for example, compaction, mineral precipitation/dissolution, and weathering, can significantly alter a rock's texture and mineralogy, resulting in the change in k (Cardell et al., 2008; Noiriel et al., 2004; Popp et al., 2001). To better understand the fluid flow phenomenon in rocks, it is therefore necessary to analyze how diageneses affect a rock's microstructure and thus its permeability. Such analysis is an area of interest for studying many geological systems and processes, such as crustal hydrothermal systems (Steefel & Lasaga, 1994), fluid flow in Earth's mantle (Kelemen et al., 1995), and saline water circulating in carbonates (Whitaker & Smart, 1997). In particular, these analyses are critical for carbonate rocks, of which the permeability-porosity relations do not follow the trend observed in sandstones (e.g., Ehrenberg & Nadeau, 2005) due to more severe precipitation and (or) dissolution. From a practical perspective, the knowledge of diagenetic influences on k, especially

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those related to mineral precipitation and dissolution, could benefit many engineering projects such as geological CO_2 sequestration (Luquot & Gouze, 2009), bioremediation and soil improvement with microbially induced calcite precipitation (Achal et al., 2011; Montoya & DeJong, 2015), and oil recovery by injecting acid in reservoirs (Emberley et al., 2004).

To date, there have been many studies on the microstructural and permeability variations of rocks during mineral precipitation and/or dissolution (e.g., see a recent study in Luhmann et al., 2017, and the references therein). These studies can be grouped into two categories: laboratory experiments and pore-scale numerical simulations. Most laboratory experiments are conducted under controlled flow and environmental conditions (e.g., Garcia-Rios et al., 2015), and the samples are characterized in detail with advanced pore-scale imaging techniques (Noiriel, 2015). These experiments provide valuable data for us to understand different precipitation/dissolution patterns in rocks and their controlling factors. As an alternative, a pore-scale numerical simulation has also been widely used in studying the effect of precipitation/dissolution on rock properties (e.g., Miller et al., 2017; Molins et al., 2017; Qin et al., 2016; Yoon et al., 2012). One advantage of numerical simulations is that many microscopic parameters of rocks, which are difficult to determine in experiments, can be directly extracted from the digital representation of the microstructure of the samples (Blunt et al., 2013). Another advantage is that different precipitation/dissolution patterns (e.g., uniform dissolution versus wormholing; Golfier et al., 2002) can be relatively easily controlled in the simulation (e.g., Jiang & Tsuji, 2014). Currently, the main focus of pore-scale simulations is still on developing microscale precipitation/dissolution models that can reproduce the variations in a rock's microstructure and bulk properties as observed in laboratory experiments (e.g., Miller et al., 2017; Nunes et al., 2016).

One of the most important goals in studying rocks experiencing precipitation and/or dissolution is to develop a predictive k model. Currently, a large body of the relevant research centers on empirical porosity-permeability models such as the power law model and Kozeny-Carman equation (e.g., Bernabé et al., 2003; Zhu et al., 2007). The general conclusion from these studies is that the model parameter(s) varies considerably during the diagenetic processes. Therefore, the permeability predictions from these empirical relations with fixed model parameters are not satisfactory. Recently, attempts have been made to incorporate diagenetic processes in the Kozeny-Carman equation (e.g., Ghezzehei, 2012), and these process-based studies have provided valuable, pore-scale insights into the diagenetic controls on rock's permeability.

At laboratory scales, the permeability of rocks can also be estimated with the aid of geophysical data such as electrical resistivity (Revil & Cathles, 1999; Slater & Lesmes, 2002), induced polarization (Binley et al., 2005; Revil et al., 2015), and nuclear magnetic resonance (NMR) relaxation (Daigle & Dugan, 2011; Kenyon et al., 1988). The underlying principle of these applications is the use of geophysical parameters (i.e., formation factor, imaginary conductivity, induced polarization relaxation frequency, and NMR $T2$ relaxation time) to estimate the hydraulic tortuosity (Duda et al., 2011) or the effective pore radius. By doing so, the permeability of the rock can then be estimated from a simple torturous bundle of capillary tubes model (e.g., Cai et al., 2014) without knowing the detailed rock texture (see detailed discussions in section 2). Due to the noninvasive nature and applicability in field tests, predicting permeability from geophysical data could potentially be a useful means to determine the spatial hydraulic heterogeneity in field scales (e.g., Maurya et al., 2018) if the lab-derived petrophysical relations can be properly upscaled to field scales. Despite the existence of many geophysical-based k prediction models (see Binley et al., 2015), it is, however, unclear whether they still hold for rocks experiencing different patterns of precipitation or dissolution.

This study aims to understand whether the commonly used, electrical tortuosity-based permeability model, $k = r_{\mathrm{eff}}{}^2/8F$ (r_{eff} is an effective pore size, and F is the formation factor; e.g., see Avellaneda & Torquato, 1991), still works for rocks in different mineral precipitation/dissolution scenarios using a pore-scale, numerical approach. In section 2, the theoretical background of the permeability model is reviewed. Section 3 describes the details of the pore-scale methods used to simulate precipitation/dissolution, electric current flow, and fluid flow in rocks. The results, including permeability and formation factor, are presented in section 4. In section 5, permeability estimated from $k = r_{\mathrm{eff}}{}^2/8F$ is compared with the hydrodynamic calculations and the origins of the prediction errors are also discussed.

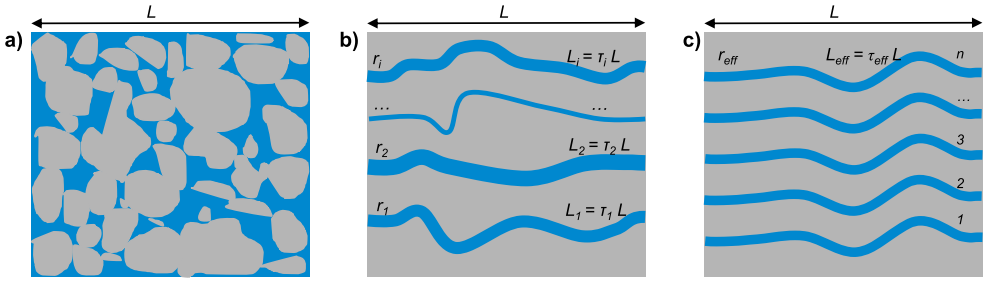


Figure 1. Schematics demonstrating different models of a porous rock (pores are indicated in blue, and rock matrix is indicated in gray): (a) a real porous medium, (b) a simplified model consisting of torturous circular tubes with different size r_i and length L_i, and (c) a simplified model having many torturous circular tubes with same size r_{eff} and length L_{eff}. The length of the rock sample is L, and the related k expression is also shown in each figure. In Figure 1a, only the connected pores are considered for simplifications. The porosities in Figures 1b and 1c are identical, and they may be less than the actual porosity shown in Figure 1a.

2. Permeability Prediction—Theoretical Background

The permeability k of a porous medium (with a dimensional unit of $[\text{length}]^2$) can be determined experimentally by measuring the flow rate Q induced by a pressure difference ΔP across the sample with Darcy's law (e.g., Lu & Likos, 2004), that is,

$$k = \frac{Q\mu}{\Delta PA}, \tag{1}$$

where μ is the viscosity of the fluid and A is the cross-section area of the sample. According to dimensional analysis, the permeability k is a product of a squared length scale, that is, l^2, and a dimensionless and scale-invariant quantity H (Bernabé et al., 2010):

$$k = Hl^2. \tag{2}$$

Based on equation (2), Bernabé et al. (2010) pointed out that the modeling of k essentially consists of two steps: (1) the determination of the relevant length scale l and (2) the selecting of the scale-invariant, dimensionless parameter H. This viewpoint can be used to understand most, if not all, of the existing permeability prediction models including those incorporating geophysical measurements as will be discussed in details later.

The pore space of a real porous rock usually has a complex structure as shown in Figure 1a, and it is extremely difficult to mathematically describe its geometry (Torquato, 2002). In order to theoretically model the fluid flow, the complex pore space is usually represented by a simplified, equivalent pore space. Here "equivalent" means the permeability of porous media with these two pore spaces is the same. Generally, only connected pores of the rock are considered in the simplification because isolated pores usually do not contribute to the fluid flow. One commonly used model is a bundle of tortuous circular capillary tubes with a dispersed radius r_i and length L_i as shown in Figure 1b. In this simplified model, the tubes are in parallel and the fluid flow in each tube can be described by the Hagen-Poiseuille equation. The analytical form of k for the model in Figure 1b can then be derived as

$$k = \frac{1}{8A}\sum\frac{\pi r_i^4}{\tau_i} \tag{3}$$

where $\tau_i = \frac{L_i}{L}$ is the tortuosity of tube i (L being the length of the sample).

Many permeability models, especially for unsaturated porous media, are based on equation (3), and popular examples include those proposed in Burdine (1953), Mualem (1976), Fredlund et al. (1994), and Kosugi (1996). Note that this type of model needs the pore size r_i distribution as an input. In practice, the pore size distribution of soils is usually estimated from the water retention curve or particle size distribution, and for rocks, it is usually measured with mercury injection capillary pressure curves. Also, since the tortuosity τ_i

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distribution is not available, the average tortuosity $<\tau_i>$ is commonly used instead of the distribution. In practice, $<\tau_i>$ can be assumed as unity if a bundle of straight capillary tubes model is used (e.g., Kosugi, 1996) or a nonunity constant (e.g., $<\tau_i> = \sqrt{2}$ for sphere packing in Carman, 1937). More commonly, $<\tau_i>$ is assumed as a function of porosity ϕ (for saturated cases) or water content θ (for unsaturated cases); the parameters in these functions are sometimes constrained by texture data (Schuh & Cline, 1990) or geophysical data (e.g., resistivity data; Niu et al., 2015).

The complexity of the pore space in Figure 1b can be further reduced by assuming that r_i and τ_i are constant (i.e., effective radius r_{eff} and effective tortuosity τ_{eff} as shown in Figure 1c) and keeping the (connected) porosity unchanged. Similarly, the permeability k can be determined analytically by applying the Hagen-Poiseuille equation and Darcy's law, expressed as

$$k = \frac{\phi r_{\text{eff}}^2}{8\tau_{\text{eff}}}. \tag{4}$$

In practice, the effective tortuosity τ_{eff} is often estimated from resistivity measurements. Ignoring the surface conduction (Revil & Glover, 1998), the electrical conductivity σ_{eff} of the sample in Figure 1c is expressed as

$$\sigma_{\text{eff}} = \frac{n\pi r^2}{A\tau_{\text{eff}}}\sigma_{\text{f}} = \frac{\phi}{\tau_{\text{eff}}}\sigma_{\text{f}}, \tag{5}$$

in which n is the total number of tubes and σ_{f} is the fluid conductivity. Applying the definition of formation factor $F = \sigma_{\text{f}}/\sigma_{\text{eff}}$ (Archie, 1942), we can have $\tau_{\text{eff}} = F\phi$ for the model in Figure 1c. Thus, equation (4) can be rewritten as

$$k = \frac{r_{\text{eff}}^2}{8F}. \tag{6}$$

The above equation is the basis of using resistivity measurements to predict k. The underlying assumption is that the electrical tortuosity ($F\phi$) is the same as the geometric/hydraulic tortuosity τ_{eff}. In this paper, we term equation (6) as the electrical tortuosity-based permeability model. Compared to the dimensional analysis in equation (2), the term $1/(8F)$ is then the invariant parameter H.

Now, the next step is to estimate the effective pore size r_{eff} (i.e., the relevant length scale l in equation (2)). The most common way is to use the hydraulic radius r_{h} defined as

$$r_{\text{h}} = \frac{2}{S_{\text{por}}} = \frac{2V}{S}, \tag{7}$$

where V and S are respectively the volume and surface area of the pore space and S_{por} is the surface area-to-pore volume ratio. Both experimental and theoretical studies have indicated strong links between S_{por} and some geophysical responses, such as quadrature conductivity σ'' (Binley et al., 2005; Börner, 1992; Weller & Slater, 2015) and NMR $T2$ relaxation time (Borgia et al., 1996). This is the reason that σ'' and NMR $T2$ time have been frequently used to aid the k prediction in porous media (Daigle & Dugan, 2011; Revil & Florsch, 2010; Weller et al., 2015). Note that other factors may also affect the relations between geophysical responses and S_{por}. For instance, the change of salinity and frequency could significantly affect the value of σ'' (Weller et al., 2011); the existence of inhomogeneous magnetic field and bulk relaxation could also pose difficulties in interpreting the NMR $T2$ relaxation time (Behroozmand et al., 2015).

The dynamic pore size Λ proposed in Johnson et al. (1986) has also been used to estimate r_{eff} in k predictions. For porous media, this parameter describes the effects of the internal surface on many processes such as electrical surface conduction and high-frequency viscous damping of acoustic waves (Johnson et al., 1986). The parameter Λ is defined as

$$\Lambda = \frac{2\int |\mathbf{E}|^2 \mathrm{d}V}{\int |\mathbf{E}|^2 \mathrm{d}S}, \tag{8}$$

where $\mathbf{E}$ is the local electrical field and $\mathrm{d}V$ and $\mathrm{d}S$ denote the integrations are over the pore space and grain-fluid interface, respectively. The Λ parameter is a dynamic length that is determined from the solution of the Laplace equation, and it cannot be measured by a simple geometrical analysis (Kostek et al., 1992). In practice, Λ of a sample can be estimated through experiments, for example, by relating Λ to the characteristic

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length scale l_c (Banavar & Johnson, 1987), which corresponds to the threshold pressure p_c at which the mercury injection curve starts rapidly rising (Katz & Thompson, 1986).

In pore-scale imaging and modeling, some other pore size parameters are also frequently used, including the pore node/body size r_d and pore throat size r_{th} (Lindquist et al., 2000). With the recent advancements in imaging techniques (Wildenschild & Sheppard, 2013), extracting these geometrical parameters from microscopic images becomes readily available, and they have been commonly used to reconstruct the rock structure and to predict the permeability (Blunt et al., 2013).

In summary, equation (6) will be tested in this study with the effective pore size r_{eff} approximated by four pore size parameters, r_h, Λ, r_d, and r_{th}. Then, we have the following four electrical tortuosity-based permeability models to analyze:

$$k = \frac{r_h^2}{8F}, \tag{9a}$$

$$k = \frac{\Lambda^2}{8F}, \tag{9b}$$

$$k = \frac{r_d^2}{8F}, \tag{9c}$$

$$k = \frac{r_{th}^2}{8F}. \tag{9d}$$

3. Numerical Methods and Samples

In this section, we introduce the pore-scale, numerical methods used to simulate mineral precipitation/dissolution, fluid flow, and electric current flow in porous media. We also explain briefly how the pore size parameters (r_h, Λ, r_d, and r_{th}) are determined from digital representations of the rock microstructure. These numerical methods are applied on two rock samples, and the properties of the samples are described lastly. Note that all the simulations in this study are performed on three-dimensional samples.

3.1. Precipitation/Dissolution Patterns and Their Modeling

When reactive fluids flow through rocks, due to fluid-rock disequilibrium, chemical reactions such as mineral precipitation/dissolution could happen at the solid-liquid interface (e.g., Morse et al., 2007). This pore-scale chemical reaction could alter a rock's microstructure and thus change the rock's bulk property such as permeability. It is generally believed that the reaction rate is quite heterogeneous within the rock and a number of factors could affect the precipitation/dissolution pattern, including local fluid chemistry, mineralogy, and transport properties (e.g., Molins et al., 2017). Thus, a direct simulation of the precipitation/dissolution at the pore scale (e.g., Nunes et al., 2016) is computationally costly and requires the solution of the advection-diffusion-reaction equation.

Fortunately, the main regimes of precipitation and dissolution have been identified based on experimental and numerical studies (e.g., Golfier et al., 2002; Kang et al., 2003; Nunes et al., 2016), and they can be characterized by the combination of Péclet and Damköhler numbers, that is, *Pe* and *Da*. The *Pe* number compares the rate of advection to the rate of diffusion (e.g., see Kang et al., 2003), and it can be defined as $Pe = uL_c/D_m$ where u and L_c respectively are characteristic Darcy velocity and length of the system and D_m is the diffusion coefficient. The *Da* number is defined as the ratio of advective time over reaction time (e.g., Nunes et al., 2016), that is, $Da = k_r/u$, where k_r is the local reaction rate. In this study, we use simplified models to mimic the precipitation and dissolution at the pore scale, aiming to capture the main macroscopic features of the samples as observed in laboratory experiments. In the simulation, the fluid transport is assumed as advection dominated, that is, $Pe > 1$. Under this transport condition, two limiting cases will be considered (Nunes et al., 2016): (1) the transport-limited case, in which the reaction at the solid-liquid interface is limited by the diffusion of reactants to and from solid surfaces, that is, $PeDa < 1$; and (2) the

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reaction-limited case, in which the reaction is limited by the reaction rate at the solid-liquid interface (i.e., $PeDa > 1$). In carbonate dissolution, the transport-limited case is typically related to the "wormholing," while the reaction-limited case generally corresponds to the so-called uniform dissolution phenomena (Golfier et al., 2002).

As observed in laboratory experiments, the reaction-limited precipitation/dissolution pattern (defined at the sample scale) is characterized by a roughly uniform growth/removal of minerals at the solid-liquid interface (e.g., Noiriel et al., 2016). Here we use a geometry-based model to simulate the reaction-limited precipitation and dissolution. This model is adopted from the random growth model used in Noiriel et al. (2016) to simulate crystal growth on aragonite grains. The mineral crystals are added (in precipitation) or removed (in dissolution) through a series of discrete steps, and the following summarizes the procedures to realize it in our simulations. First, the voxels at the solid-liquid interface are identified, and this can be done by tracing the phase connectivity to identify a phase change (e.g., see Zhan et al., 2010). In precipitation, these voxels are within the pore space; in dissolution, they are in solid phases. Second, we convert the binary microstructure into a porosity field. By doing this, a continuous growth/erosion of the pore wall can be realized later (Miller et al., 2017). During a time step Δt, the porosity of the interface voxels is updated by a porosity change of $\Delta\theta < 1$ (i.e., the porosity of the voxel is decreased by $\Delta\theta$ in precipitation or increased by $\Delta\theta$ in dissolution), and the locations of the updated voxels are randomly selected following a uniform distribution with a possibility of p. These two steps are repeated to mimic the heterogeneous mineral precipitation/dissolution at the pore scale. In this process, if the porosity of an interface voxel reaches 0 (in precipitation) or 1 (in dissolution), the phase of the voxels is then converted. The above steps are repeated on the discretized rocks until the target porosity is reached.

Different from the reaction-limited process, the transported-limited (advection-dominated) precipitation and dissolution are even more heterogeneous at the pore scale (e.g., Ott & Oedai, 2015) because the reaction is constrained by the availability of reactants transported to or from the interface, which are related to the local flow conditions. In this study, we use the strategy proposed in Miller et al. (2017) to simulate this limiting case. The method assumes the local fluid velocity field is a proxy of the solvent flux, and it is equivalent to solving the zeroth-order rate equation at the solid-liquid interface (Miller et al., 2017). Overall, the numerical realization is similar to that of the reaction-limited case except that the amount of local porosity change $\Delta\theta$ during Δt is not a constant but scaled by $v_m/v_{\max}$, where v_m is the local average fluid velocity (e.g., five-voxel neighborhood) and $v_{\max}$ is the maximum fluid velocity in the pore space. In the simulation, the fluid velocity field of the sample is determined by solving the linear Stokes equation (see section 3.2). As the pore space is continuously altered by mineral precipitation/dissolution, the fluid velocity field in the pore space in our simulation is also updated in the processes.

3.2. Fluid and Electric Current Flow Simulations

To determine the fluid velocity field in the pore space of the sample, the linear Stokes equation is numerically solved. In the calculation, we use the Stokes solver developed at the National Institute of Standards and Technology (Garboczi, 1998). After applying a potential gradient across the sample, the code solves the spatial distribution of the pressure and fluid velocity using a staggered-grid finite difference scheme. The permeability k of the sample can then be determined by applying Darcy's law based on the averaged fluid velocity (e.g., Zhan et al., 2010). The Stokes solver is written in FORTRAN, and the preprocessing code is written in C. For a sample with 300^3 voxels, the calculation usually takes about 1 to 3 hr on a personal computer depending on the complexity of the model.

The electric flow in the sample is simulated using the finite difference code DC3d developed in the National Institute of Standards and Technology (Garboczi, 1998). In the simulation, the pore fluid conductivity σ_f is kept unchanged during the mineral precipitation/dissolution. The solid phase is assumed to have a uniform material property, and its electrical conductivity is zero. The conduction contributed from the electrical double layer at the solid-liquid interface (i.e., surface conduction; Revil & Glover, 1998) is ignored, and thus, no extra conductivity is added to either fluid or solid phase in the calculation (Niu & Zhang, 2017). The assumption of zero surface conductivity corresponds to rocks at the high-salinity limit where the conduction from bulk fluid is much higher than the surface conduction. It is noted the (intrinsic) formation factor F is an intrinsic rock property, which is independent of the fluid and solid-liquid interface properties (e.g., see equation 7a in Schwartz et al., 1989). The code applies a uniform electric field across the sample and then solves

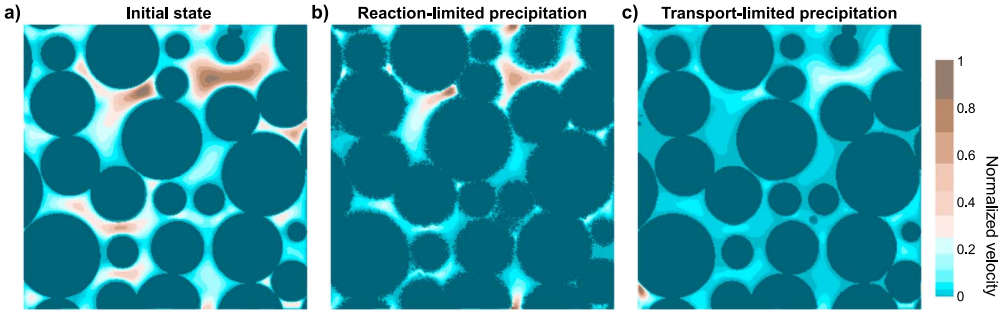


Figure 2. Microstructure and fluid velocity field of Sample 1: (a) initial condition ($\phi = 30.2\%$), (b) after reaction-limited precipitation ($\phi = 20.1\%$), and (c) after transport-limited precipitation ($\phi = 28.2\%$). The solid phase is indicated in dark blue. The fluid velocity is normalized by the maximum flow velocity in the pore space. The resolution of the images (size of each voxel) is ~4.3 μm, and the sample volume is 300^3 voxels.

the spatial electric potential and current density in the sample with the conjugate gradient method. Based on averaged current density, Ohm's law can be used to calculate the effective electrical conductivity of the sample σ_{eff} (e.g., Niu & Zhang, 2018). Since σ_{eff} is solely from the ionic conduction in the bulk solution, the formation factor F of the sample can be simply determined as $F = \sigma_f/\sigma_{eff}$. The code is also written in FORTRAN, and the calculation time usually is shorter than that of the Stokes solver.

3.3. Calculation of Pore Size Parameters

As defined in equation (7), the hydraulic radius r_h is related to the surface area S of the solid-liquid interface and the volume V of the pore space. In the simulation, the solid-liquid interface of the sample is determined by tracing the phase connectivity (e.g., see Zhan et al., 2010). For each voxel, its six neighbors are checked to identify the phase change. If a voxel is isolated with zero connectivity, we simply eliminate it. After the interface is identified, V and S can then be easily determined, and then r_h can be calculated. The determination of Λ is similar to that of r_h, but the weighting of each voxel is not unity but dependent on the electrical field $|\mathbf{E}|^2$ according to equation (8). The electrical field $\mathbf{E}$ is the gradient of the electrical potential field, which is the solution of the Laplace equation.

As discussed, the mean pore node size r_d and pore throat size r_{th} are only related to the geometry of the pore space, and they can be determined using the concept of maximum balls (e.g., Al-Kharusi & Blunt, 2007) or by medial axis analysis of the pore space (e.g., Lindquist et al., 1996). In this study, we use the modified maximal ball algorithm developed by Dong and Blunt (2009) to determine the distribution of the pore node size and pore throat size. This two-step algorithm searches the nearest solid to define a void ball, and it defines pore nodes and pore throats by affiliating the maximal balls into family trees according to their size and rank (Dong & Blunt, 2009). The results are then fitted with a normal distribution to determine the mean value r_d and r_{th}.

3.4. Samples

Two samples are used in this study: a synthetic loosely packed ooids sample (Sample 1) and a carbonate mudstone (Sample 2). The ooids sample (Figure 2a) with the initial porosity $\phi = 30.2\%$ is obtained using the discrete element method by compressing sparsely distributed spherical particles (e.g., Niu & Zhang, 2018). The carbonate mudstone sample has a porosity of ~13% (from helium measurement) and is from the Wellington Formation, KS; its microstructural image in Figure 3a is from micro-computed tomography scan data after segmentation. In this study, Sample 1 (Figure 2a) is used for precipitation simulations and Sample 2 (Figure 3a) is used for dissolution simulations.

4. Simulation Results of Fluid and Electric Current Flows

The limiting scenarios of precipitation and dissolution are numerically simulated using the methods mentioned above for Samples 1 and 2, respectively. Their permeability k and formation factor F are also calculated during the precipitation and dissolution, and the results are shown in Figures 4 and 5. To facilitate

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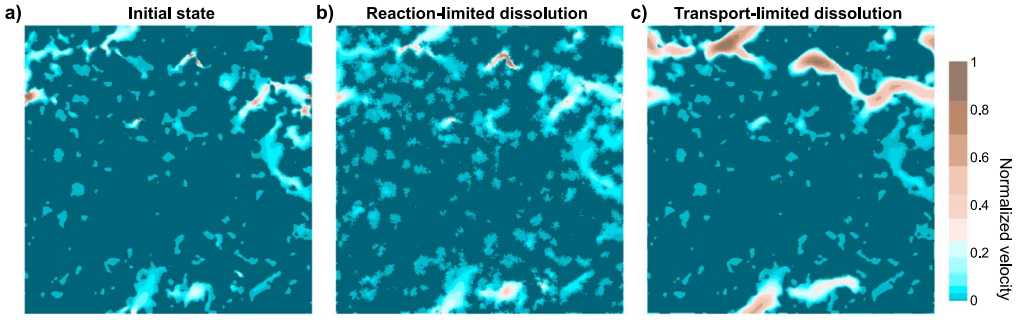


Figure 3. Microstructure and fluid velocity field of Sample 2: (a) initial state ($\phi = 13.8\%$), (b) after reaction-limited dissolution ($\phi = 28.8\%$), and (c) after transport-limited dissolution ($\phi = 17.1\%$). The solid phase is indicated in dark blue. The fluid velocity is normalized by the maximum flow velocity in the pore space. The resolution of the images (size of each voxel) is ~4.3 μm, and the sample volume is 300^3 voxels.

the discussion, we also plot the empirical power law of k (e.g., Zhu et al., 2007) and Archie's law (Archie, 1942) with varying model parameters in the figures. The two models are expressed as

$$\frac{k}{k_0} = \left(\frac{\phi}{\phi_0}\right)^n \tag{10}$$

and

$$F = \phi^{-m}, \tag{11}$$

where k_0 is the permeability of the sample at the initial porosity ϕ_0, the exponent n is an empirical constant, and Archie's porosity exponent m is usually called cementation factor. The k-ϕ and F-ϕ relations of rocks have been extensively studied with laboratory experiments (e.g., Bernard et al., 2007; Noiriel et al., 2004). In the following, we discuss whether our simulated k and F show similar trends as those in published studies and analyze their associated microstructural variations.

4.1. Precipitation (Sample 1)

In simulating the reaction-limited precipitation, the porosity change ($\Delta\theta$) of a voxel during a time step Δt is set as 0.5, and only 5% of the interface voxels are allowed to change phase; a total of 100 time steps are simulated in the calculation, and this induces a porosity reduction of ~10%, from ~30.2% to ~20.1% (Figure 4b). In simulating the transport-limited precipitation, the maximum $\Delta\theta$ value is 0.3, and 200 time steps are considered in the simulation; the resulted porosity decrease is insignificant, only ~2% as shown in Figure 4b.

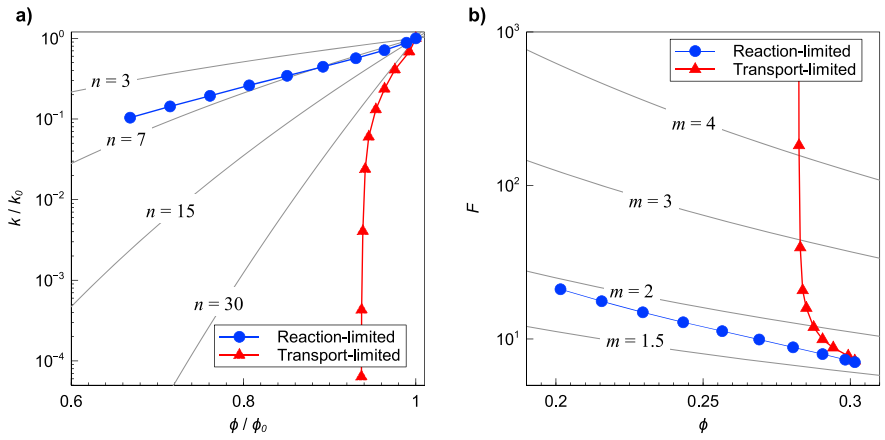


Figure 4. Simulation results of Sample 1: (a) k evolution and (b) F evolution during reaction-limited and transport-limited precipitation. The parameters n and m are the exponents in $k/k_0 = (\phi/\phi_0)^n$ and $F = \phi^{-m}$ (i.e., equations (10) and (11)).

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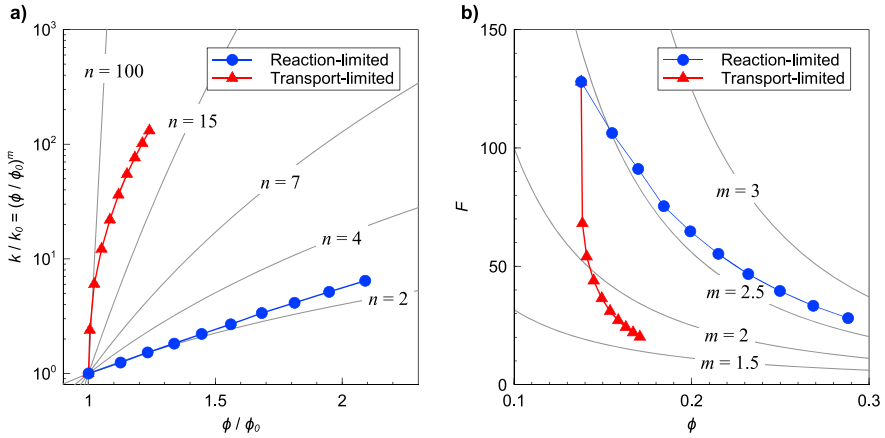


Figure 5. Simulation results of Sample 2: (a) permeability k evolution and (b) F evolution during reaction-limited and transport-limited dissolution. The parameters n and m are the exponents in $k/k_0 = (\phi/\phi_0)^n$ and $F = \phi^{-m}$ (i.e., equations (10) and (11)).

The simulated k-ϕ and F-ϕ relations are quite different for the two precipitation. For reaction-limited case (blue symbols in Figure 4), both exponents (n and m) are almost constant without a large variation. For example, in the entire porosity range, n is close to ~5, while m is roughly between 1.5 and 2. The microstructure of the sample after the reaction-limited precipitation is shown in Figure 2b, and it appears that, at the sample scale, the precipitated minerals uniformly distribute on the grain surface. These numerical features are consistent with those observed in high-porosity Fontainebleau sandstones (Bourbie & Zinszner, 1985) where the uniform silica overgrowth induces an insignificant variation in n and m (Gomez et al., 2010; Revil et al., 2014). It seems that the uniformly precipitated minerals did not significantly change the electric/fluid flow patterns in the pore space (e.g., see Figures 2a and 2b) and that the reaction-limited precipitation may be characterized by a constant porosity exponent n (or m).

For transport-limited cases in Figure 4 (red symbols), both exponents varied considerably despite the minor porosity reduction. For instance, n quickly exceeds ~30 after a porosity decrease of ~2%, and m lastly reaches ~5.1 when $\phi = 28\%$. The increased m and n in transport-limited precipitation are associated with the fact that the precipitated minerals accumulate mostly near the pore throats where fluid velocity is relatively high (see Figures 2a and 2c). These numerical results agree with the experimental observations in many published studies (e.g., see Noiriel et al., 2004; Leroy et al., 2017). Thus, it is clear that the transport-limited precipitation has a profound influence on the fluid/electric flow, and it may be characterized by a sharply increased n (or m).

4.2. Dissolution (Sample 2)

The same parameters (e.g., Δt, $\Delta\theta$, and time step) were used to simulate mineral dissolution in Sample 2, and the calculated k and F are shown in Figure 5. For reaction-limited dissolution (blue symbols), both k/k_0 and F are power functions of porosity ϕ with constant exponents, for example, $n = \sim 2$ in Figure 5a and $m = \sim 2.5$ in Figure 5b. In transport-limited cases (red symbols), however, the associated exponents (n and m) vary largely. For instance, the porosity exponent n decreases from ~100 at $\phi = 13.8\%$ to ~20 at $\phi = 17.1\%$. This huge decrease in n has been experimentally observed in limestones where a ~2% porosity increase causes n to change from ~150 to ~15 (Noiriel et al., 2004). The microstructural variations of the sample (Figure 3) are also quite similar to those of Sample 1 (Figure 2). While in reaction-limited dissolution, the dissolved minerals are almost uniformly distributed on all the solid-liquid interface (Figure 3b), the transport-limited dissolution mainly occurs in some preferred channels (Figure 3c) where the fluid velocity is high (Figure 3a). For both dissolution, the porosity exponent n and m can be used to characterize the microstructural variations in rocks.

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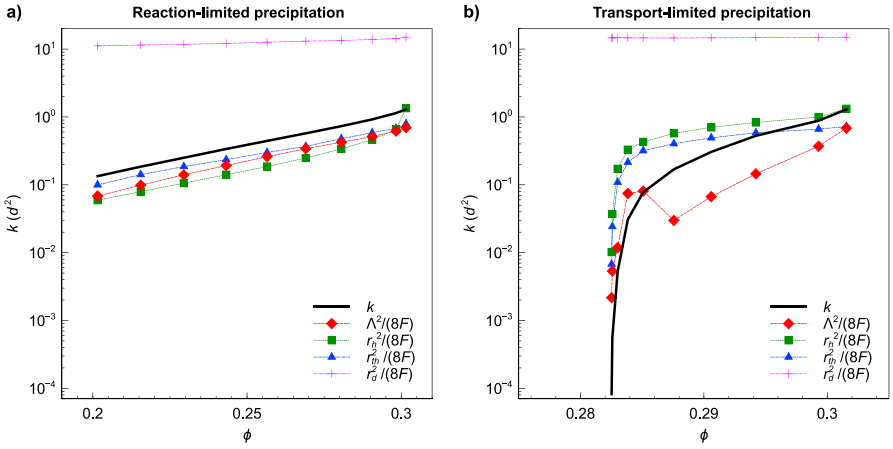


Figure 6. The predicted permeability of Sample 1 during (a) reaction-limited precipitation and (b) transport-limited precipitation. The unit of permeability k is in d^2, where d is the voxel size.

5. Permeability Predictions

In this section, we compare the k predictions to their true values for the two samples undergoing different patterns of precipitation/dissolution. We also discuss the origin of the prediction errors in each scenario. Note that our focus here is the sample's permeability in response to pore-scale precipitation/dissolution, and the results need to be upscaled to field scales for hydrogeological applications. Currently, the upscaling of lab-based, hydrogeophysical relations still remain as an unsolved problem in hydrogeology.

5.1. Permeability Predictions

The permeability predicated with (9a) is shown in Figures 6 and 7 for the two samples under different types of precipitation/dissolution. Four pore size parameters were used in the predictions, and they are Λ, r_h, r_{th}, and r_d. The permeability determined from hydrodynamic calculations, that is, solving the linear Stokes equation, is also shown for comparison. In Figures 6 and 7, one major finding is that the prediction from the pore body size r_d significantly overestimates the permeability, ~2 orders, for all the cases. This is understandable because it is the pore throat rather than pore body that controls the fluid flow in porous media.

For reaction-limited cases (Figures 6a and 7a), parameters Λ, r_h, and r_{th} give very good permeability estimates, and in general, the discrepancies are less than a half order. In contrast, in transport-limited cases (Figures 6b and 7b), all the predictions deviate noticeably from the true permeability. Among them, the pore size Λ still gives a fair k estimation, and the associated discrepancy is roughly ~0.5 order (e.g., Figure 7b).

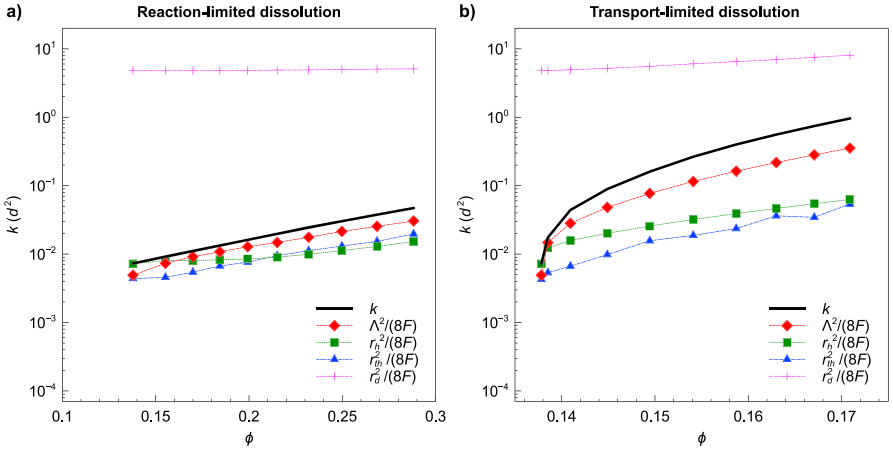


Figure 7. The predicted permeability of Sample 2 during (a) reaction-limited dissolution and (b) transport-limited dissolution. The unit of permeability k is in d^2, where d is the voxel size.

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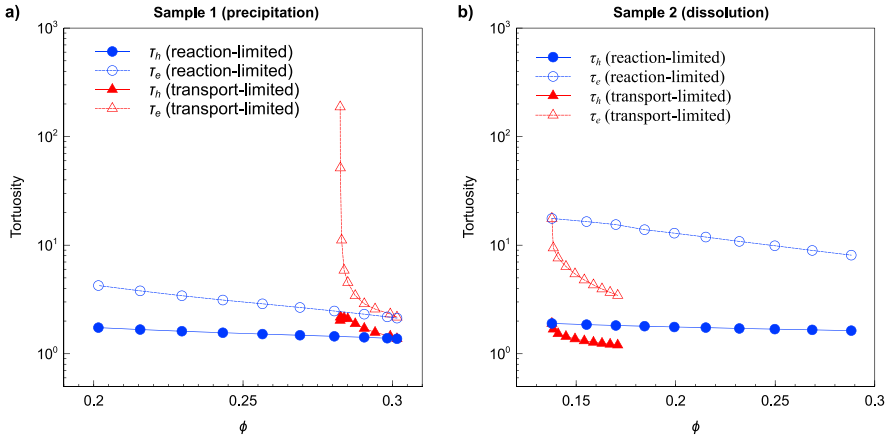


Figure 8. Evolutions of electrical tortuosity τ_e and hydraulic tortuosity τ_h during (a) precipitation (Sample 1) and (b) dissolution (Sample 2).

This indicates that the applicability of the electrical tortuosity-based permeability ((9a)) largely depends on the patterns of the mineral precipitation and dissolution. It seems the electrical tortuosity-based k prediction works better for rocks experiencing reaction-limited precipitation and dissolution where the mineral growth and removal are roughly uniformly distributed on the solid-liquid interface. Among these pore size parameters, the electrical field normalized pore size Λ works fairly well for both reaction- and transport-limited cases.

5.2. Discussion on the Origin of k Prediction Errors

According to equations (4) and (6), the uncertainty of k predictions may come from two sources: (1) the length scale r_{eff} estimation and (2) the tortuosity τ_{eff} estimation. In equation (6), while different pore size parameters (Λ, r_h, r_{th}, and r_d) are used to estimate r_{eff}, the tortuosity is estimated by the electrical tortuosity $\tau_e = F\phi$. In the following, we compare the electrical tortuosity τ_e to the true tortuosity, that is, the hydraulic tortuosity τ_h, which can be determined from the fluid velocity field, expressed as (Duda et al., 2011)

$$\tau_h = \frac{\langle v_{mag} \rangle}{\langle v_{\parallel} \rangle}, \tag{12}$$

where $\langle v_{mag} \rangle$ is the pore volume-averaged fluid velocity magnitude and $\langle v_{\parallel} \rangle$ is the pore volume-averaged fluid velocity parallel to the applied pressure gradient.

The tortuosity of Samples 1 and 2 is presented in Figures 8a and 8b, respectively. It is shown that for both samples, the calculated electrical tortuosity τ_e (empty symbols) is larger than the hydraulic tortuosity τ_h (solid symbols) in all the scenarios. In Figure 8, the ratio τ_e/τ_h is roughly ~3 for Sample 1 and ~10 for Sample 2. They correspond to an underestimation of k of ~0.3 and ~1 order. These magnitudes are in accordance with the discrepancies between predicted and true permeability as observed in Figures 6 and 7. This implies that using $F\phi$ to represent τ_{eff} could underestimate the permeability significantly. Moreover, in reaction-limited cases (blue symbols), both hydraulic and electrical tortuosity changes smoothly. In contrast, the tortuosity in transport-limited cases (red symbols) varies considerably despite the minor porosity change. These distinct responses may be associated with the different microstructural variations as discussed before (e.g., a relatively uniform change of the solid-liquid interface in Figures 2b and 3b and an increased mineral growth/removal near the main flow paths in Figures 2c and 3c).

Now we analyze the accuracy of using Λ, r_h, r_{th}, and r_d to estimate the effective pore size r_{eff} in calculating k (equation (4)). We calculate the true r_{eff} value from the true permeability k and hydraulic tortuosity τ_h, expressed as

$$r_{eff} = \sqrt{\frac{8k\tau_h}{\phi}}. \tag{13}$$

The evolutions of these pore size parameters are shown in Figures 9 and 10 for Sample 1 (precipitation) and Sample 2 (dissolution), respectively. In Figures 9a and 10a, it appears that Λ, r_h, and r_{th} are quite close to r_{eff},

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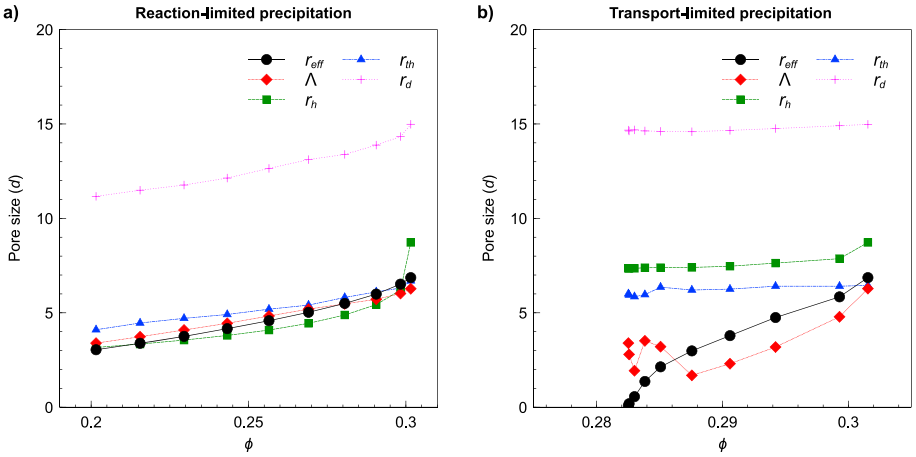


Figure 9. Evolutions of different pore size parameters of Sample 1 during (a) reaction-limited precipitation and (b) transport-limited precipitation.

indicating that the length scale controlling fluid flow can be safely estimated by any of them for reaction-limited precipitation/dissolution. This also confirms that the errors of k predictions in Figures 6a and 7a are mainly from the tortuosity estimation. In transport-limited cases, pore size parameters such as r_h and r_{th} cannot capture the significant variations of the effective pore size constraining the fluid flow, and thus, they have partially contributed to the discrepancies of k observed in Figures 6b and 7b. If compared with the tortuosity estimation (i.e., $F\phi$), the contributions of effective pore size estimations are very minor. It is also clear from Figures 9 and 10 that among these four pore size parameters, only Λ can be reliably used to estimate the length scale characterizing the fluid flow in porous media undergoing different types of precipitation and dissolution.

As shown in Figure 9, the calculated electrical tortuosity is always higher than the hydraulic tortuosity. This is because the shear plane (within which the fluid is not moving) is always away from the solid-fluid boundary. Thus, the fluid does not need to move exactly along the rough solid-fluid boundary. In contrast, ions can move within the shear plane. This means the ions near the solid-liquid interface have to move along the rough boundary surface, resulting in a longer flow path than the fluid. Therefore, the electrical tortuosity is higher than the hydraulic tortuosity as shown in Figure 9. The results imply that we have to be very cautious when using electrical tortuosity to approximate hydraulic tortuosity, especially when the internal surface of the rock is very rough.

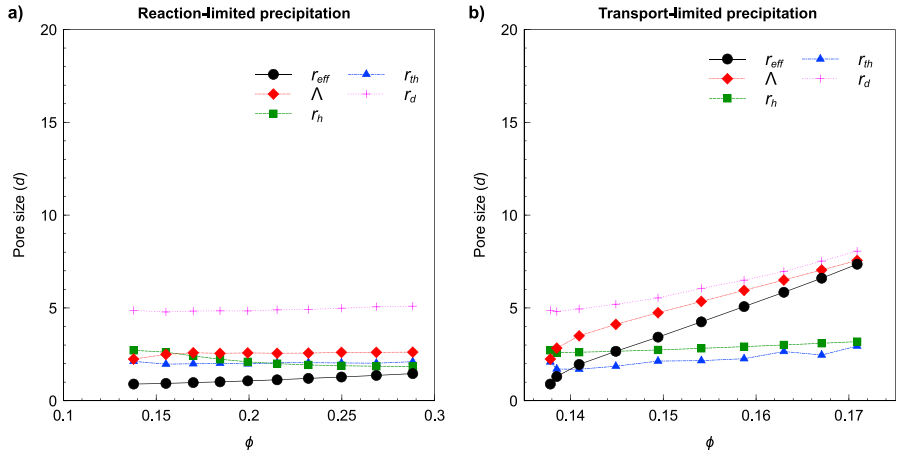


Figure 10. Evolutions of different pore size parameters of Sample 2 during (a) reaction-limited dissolution and (b) transport-limited dissolution.

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6. Conclusions

The pore-scale numerical simulations have shown that the permeability-porosity (k-ϕ) and formation factor-porosity (F-ϕ) relations of rocks are largely influenced by the patterns of the mineral precipitation and dissolution. In reaction-limited, advection-dominated cases, k and F are roughly power functions of porosity ϕ with the exponents (n and m) nearly constant, and in general, the precipitated or dissolved minerals uniformly distribute at the solid-fluid interface. This microstructural alternation is responsible for the porosity change but did not significantly change the characteristics of the fluid and electric current flows in the pore space. In transport-limited, advection-dominated cases, the precipitated and dissolved minerals are mostly near the pore throats where fluid velocity is relatively high. This heterogeneous microstructural alternation could significantly affect the fluid and electric current flows in the pore space, resulting in a huge change in k and F despite a minor variation in ϕ. The commonly used power laws with constant porosity exponents are not able to describe such changes. Compared to reaction-limited cases, transport-limited precipitation/dissolution has a more profound influence on fluid/electric flow and it may be characterized by a sharply changed n or m.

In general, the electrical tortuosity-based permeability model $k = r_{\text{eff}}^2/8F$ works well for rocks experiencing precipitation and dissolution if the effective pore size r_{eff} is estimated with the electrical field normalized pore size Λ. The error of the prediction is mainly from the electrically estimated tortuosity, that is, $F\phi$, which is consistently larger than the true (hydraulic) tortuosity. In reaction-limited cases, the hydraulic radius r_{h} and mean pore throat size r_{th} can also be safely used to estimate r_{eff}. In all the cases, the pore body size r_{d} significantly overestimates the effective pore size, and thus, it should be avoided in permeability predictions of rocks experiencing precipitation and dissolution.

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